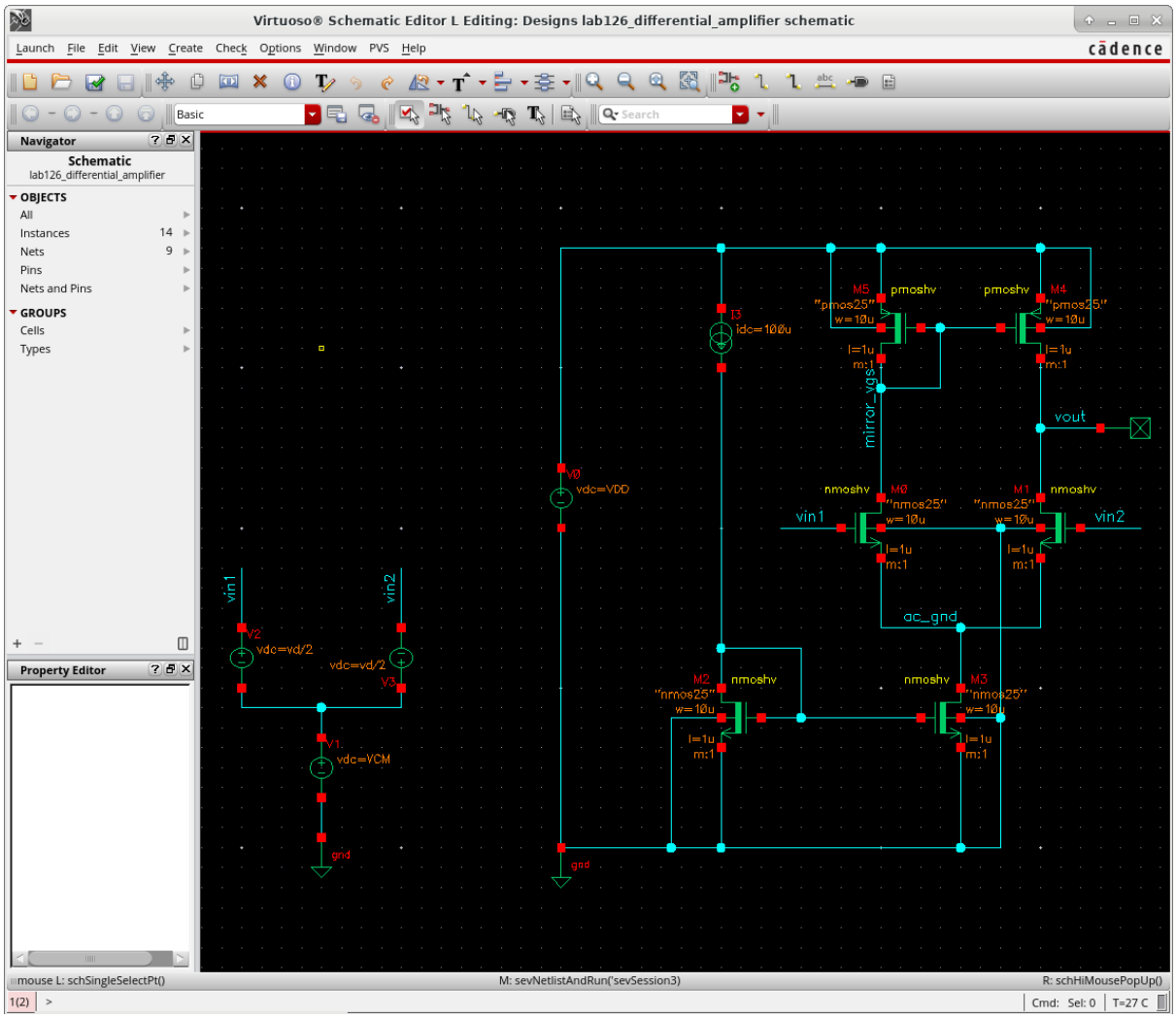
Analogue Integrated Circuits
EEMEM0006
Laboratory Instructions

The digital revolution is constructed on top of an analogue reality.



Tutorial Lab 12.1 (Formatively Assessed)

Setup, Package Introduction, Schematic Entry, and Simulation

Preliminary Matter: Setting up the Cadence Environment

Please read through sections 1 through 3 of the tutorial lab 12.1 instruction booklet/handout before entering these commands into your terminal.

For this course, you will use the Cadence Virtuoso Design Environment (referred to simply as Cadence). Cadence is a Linux based software suite for designing full custom integrated circuits. In order to use Cadence you will first need to set-up the Cadence environment.

1. Open a terminal window and ensure that you are in your home directory using the following command:

```
cd ~
```

Note: 'cd' stands for change directory. The solitary tilde symbol (~) is a shortcut/alias for your home directory (/home/username).

2. Under your home directory, use the following commands to load the Cadence environment:

```
module use /eda/cadence/modules
module load course/EENGM6030
```

Note: The two 'module' commands load the required shell/Cadence environmental variables for this course (paths, names, aliases…) In particular, this includes a reference to the process design kit (PDK) used in this course, the GPDK180. You will need to do this every time you open a terminal, since the sourced environment only applies to the current terminal (until it's closed).

3. The first time you use Cadence, you will need to run the following command (script):

```
EENGM6030_setup
```

*Note: **This step only needs to be completed ONCE - the first time you run Cadence.** This will create a folder called AICD (short for Analogue Integrated Circuit Design) under your home directory. The AICD folder contains the files and directories needed to run Cadence. Running this setup script more than once will result in a warning message that the AICD folder already exists. It will then exit without making changes.*

4. The AICD folder will be used for all your Cadence related activities for this course. **You should always make sure you are in your AICD folder before you launch Cadence.** To run cadence, use the following commands:

```
cd ~/AICD
virtuoso &
```

1. Learning Objectives of this Lab:

With guidance, you will:

- Attain a working understanding of the environment for and basic functionality of the Cadence Design Framework for analogue integrated circuit design.

These labs are designed to illustrate the principles and practices of using a schematic entry-driven, SPICE-based* simulation program to develop analogue integrated circuits. They will cover the most important steps in the analogue IC design flow.

We will use the Cadence Virtuoso Design Environment for this course, but the principle and application is equally applicable to all contemporary platforms offering an analogue IC design flow. The underlying algorithms of simulation and the measurement techniques are generic.

What these tutorial labs are not! They are not intended as a detailed treatment of device physics or circuit analysis/design. -

*SPICE - **S**imulation **P**rogram with **I**ntegrated **C**ircuit **E**mphasis.

2. Cadence Documentation, Help, and Licensing

The Cadence Design Environment is a collection of tools – Schematic, Layout, and Text Editors, Analogue Design Environment, Library Manager, Spectre Simulation Engine, and more.

PLEASE NOTE: WHEN LAUNCHING A TOOL IN THE CADENCE DESIGN ENVIRONMENT YOU MAY SOMETIMES GET A WARNING ABOUT AN UNAVAILABLE LICENSE FOR THE TOOL, AND ASKING PERMISSION TO OBTAIN A HIGHER-TIERED LICENSE. YOU SHOULD SELECT 'ALWAYS' TO THESE MESSAGES.
If you neglect to do this you can recover from the situation by selecting the *options* tab, selecting *cds environment editor* selecting *licence* searching for ADE and change the setting to *Always*.

Each tool has its own help menu. You can explore the documentation for each tool at **Menu: Help → User Guide**. While this course is not about teaching you the specifics of Cadence, you do need to be familiar with it. We do not cover all aspects of the Cadence Virtuoso toolset. There is a wealth of local and online documentation available from the help menus, covering every aspect of the Virtuoso Design Environment. You should learn to consult this documentation if you want to know more about a specific tool. (You can also access the user guides for a specific tool by pressing F1 when a particular window, dialog box, or form is in focus – though this doesn't always work).

Alternatively, you can type 'cdnshelp &' at the command prompt (assuming you have setup your Cadence environment). This will take you to the home page of the Cadence help pages. Clicking on the 'Show Library' icon will display the contents tree for all sections.

3. Getting Started with Cadence:

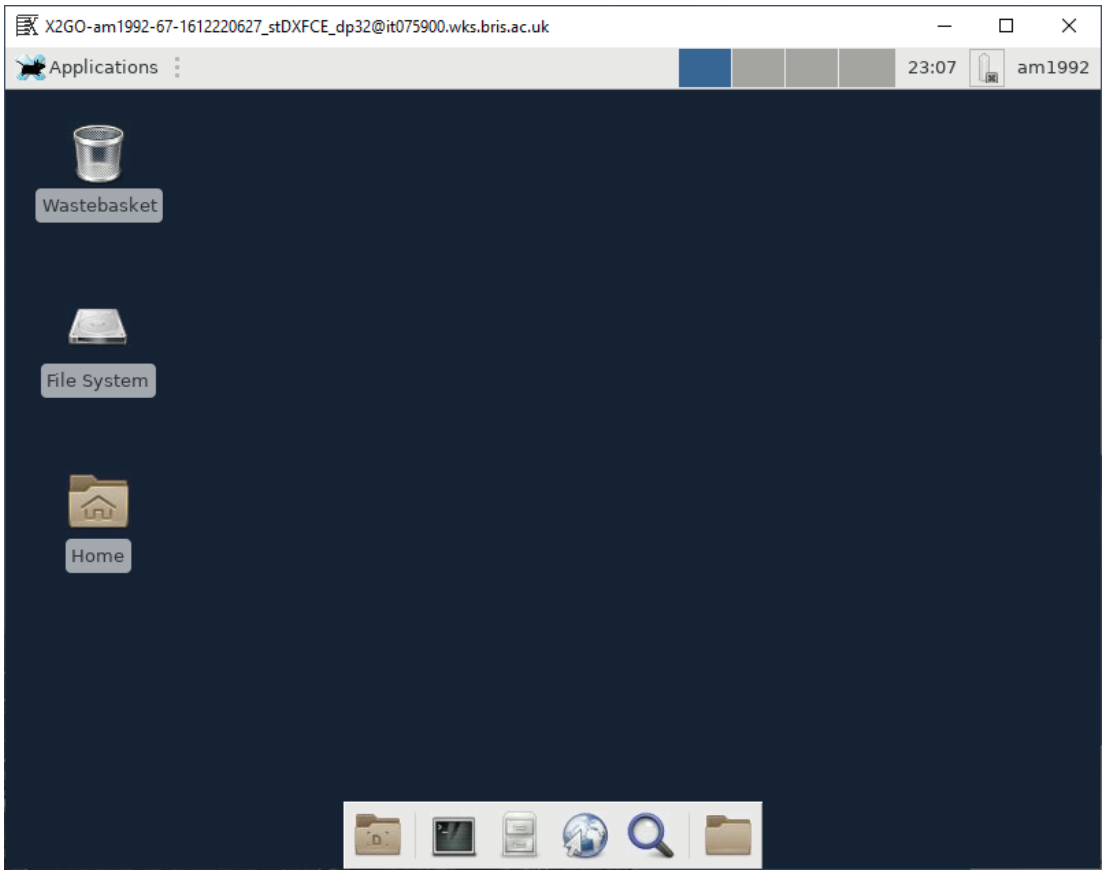
3.1 Remote Access using X2Go

(3.1.1) You may be using Cadence remotely, so you will need to log into one of the engineering Linux Lab machines (in MVB1.15 or MVB2.11) using a remote desktop client.

(3.1.2) The X2Go client runs on Windows, Mac, and Linux and provides a bandwidth efficient graphical user interface (GUI) to a remote Linux Lab machine. While you may use whichever remote desktop software you prefer (where it is compatible with the University's systems), we recommend you use X2go, especially if you are off campus. IT services provide a support page for X2go at:

https://uob.sharepoint.com/sites/itservices/SitePages/fits-engineering-linux-x2go.aspx

(3.1.3) Launch the OS. To start a remote Linux session via X2go, follow the instructions on the IT services page linked-to above (see 'Making a New Session'). When you have successfully started a remote session, you will be logged into one of the Linux Lab machines (randomly chosen from the pool of available machines). You should see your Linux desktop as in fig. 1.

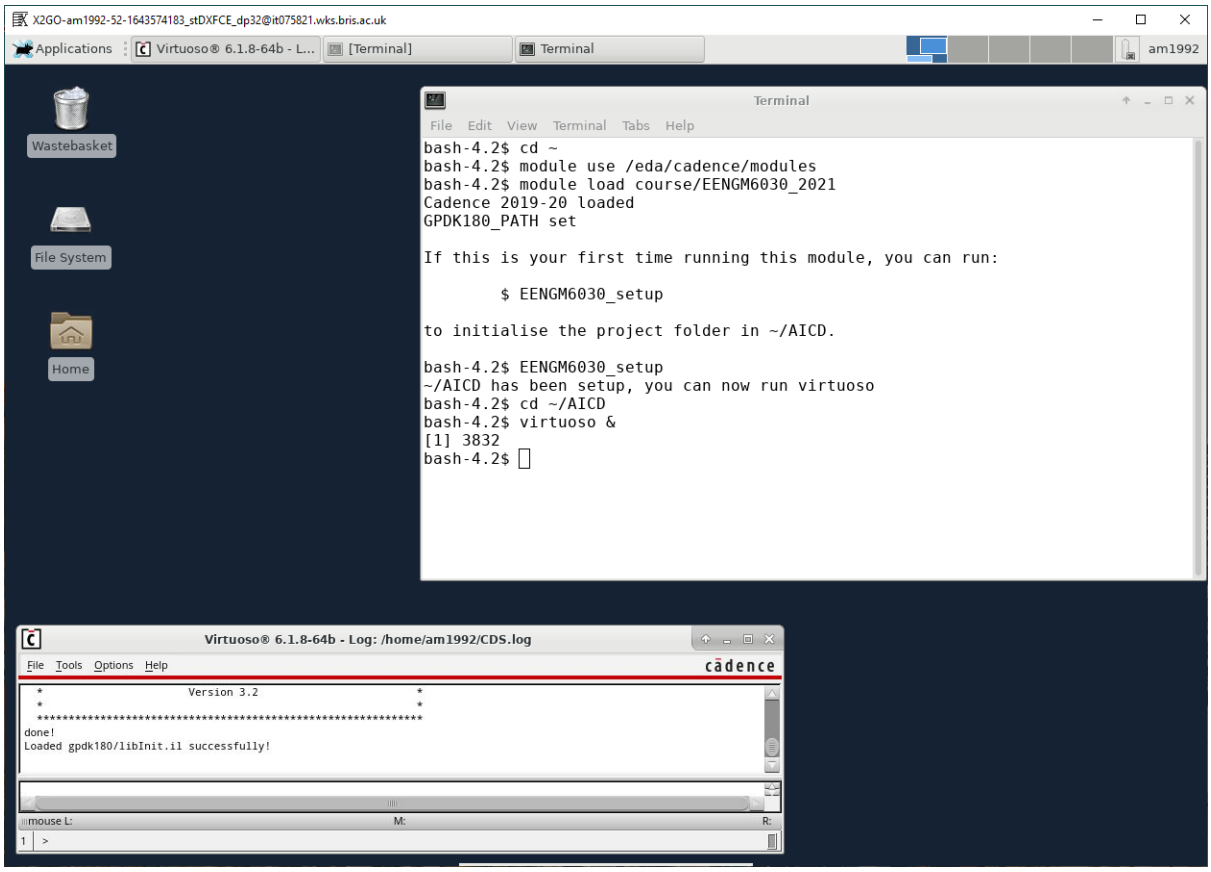


< Figure 1: Linux Desktop in an X2go window. Yours may look slightly different depending on the content of your desktop or your chosen desktop personalisations. >

3.2 Launching Cadence

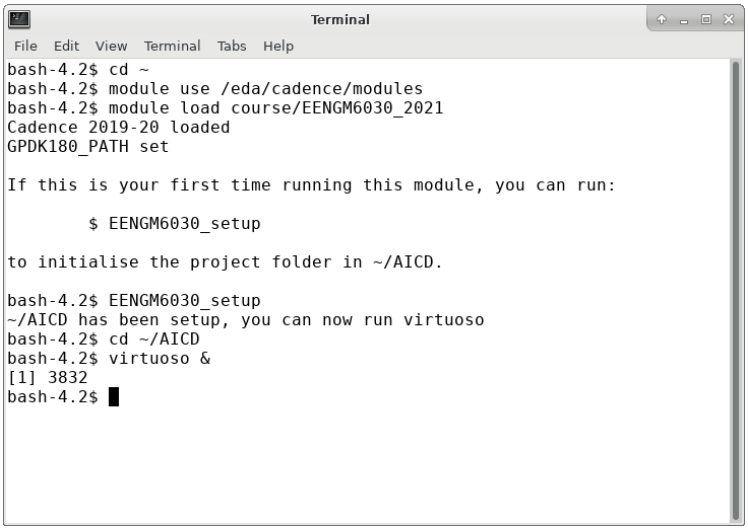
(3.2.1) In order to use Cadence, you will first need to set-up the Cadence environment. Open a terminal window (from either the launcher panel, the applications menu, or by right-clicking anywhere on the empty desktop space) and enter the commands in the order given on the 'Setting up the Cadence Environment' page at the front of this document (preliminary matter). The purpose of each command is explained in the italic text beneath each command.

(3.2.2) You will need to follow these instructions each time you start Cadence (for this course). The setup script 'EENGM6030_setup' in step (3) is used to initialise the Cadence environment before first use. In particular, it will automatically create a folder named 'AICD' directly under your home directory. The AICD folder holds all your Cadence related data for this course. **You should always make sure you are in your AICD folder before you launch Cadence. Note that you only need to run the EENGM6030_setup script the first time you run Cadence.**



< Figure 2: The Terminal and Command Interpreter Window (CIW). >

(3.2.3) Launch the Cadence Command Interpreter Window (CIW). Once you have successfully executed the commands on the 'Setting up the Cadence Environment' page, after a brief period, the Command Interpreter Window (CIW) will appear, usually on the bottom left of the screen. Depending on network activity, it may take a few minutes for the CIW to load.



< Figure 3: The Terminal Window. Setting-up and starting Cadence. Note: bash-4.2$ is the prompt for the terminal's bash shell. Do NOT type this in! >

3.3 The Command Interpreter Window (CIW)

(3.3.1) The CIW controls your design session. Take some time to explore the CIW functionality. The principal components of the CIW are

1. The **title bar.** This contains the following information:

    - The name/version of the workbench you are running (e.g. Virtuoso 6.18-64b)

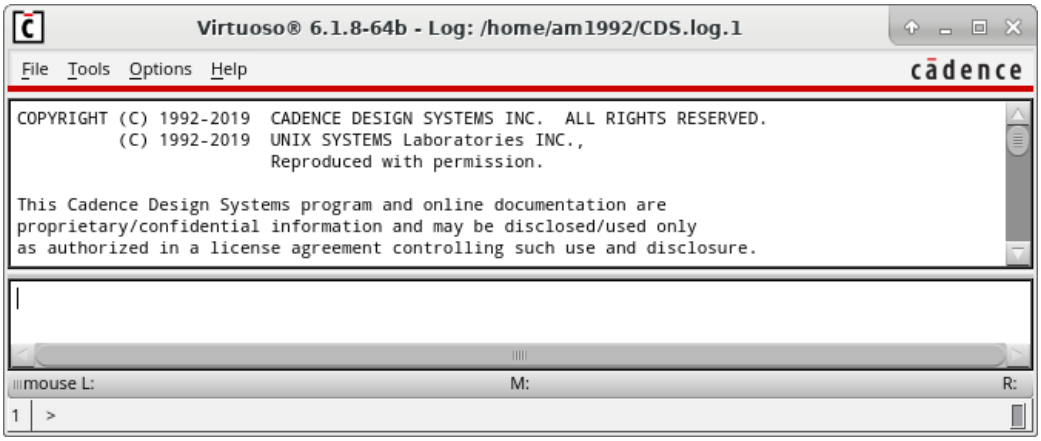
    - The path to the log file (CDS.log) that records the ongoing events of the design session. This is typically /home/username/CDS.log. The content of the log file is displayed in the output area.

    (Aside: if you used Cadence for the COMS30026 Design Verification course, the workbench would have been NCsim not Virtuoso. This would have been reflected in the CIW title bar!)

2. The **menu bar**. The items along the menu bar can change depending on any extensions used (e.g., PDKs), but the base menu items are **File**, **Tools**, **Options**, and **Help**. These are used to launch/start the various components and activities (e.g., Library Manager) of the design environment session. It is also where the online help is found.

3. The **output area**. This is the most important aspect of the CIW for our purposes. This area displays the ongoing events of the design session including command history, responses, warnings, and error messages. This is where you will come to diagnose issues/problems with your design/simulations. As well as displaying this information in the output area, the software also writes it to the CDS.log file.

4. The **edit area**. You can type a command in the edit area of the CIW below the displayed history (output area). This area is useful for running single commands or scripts.

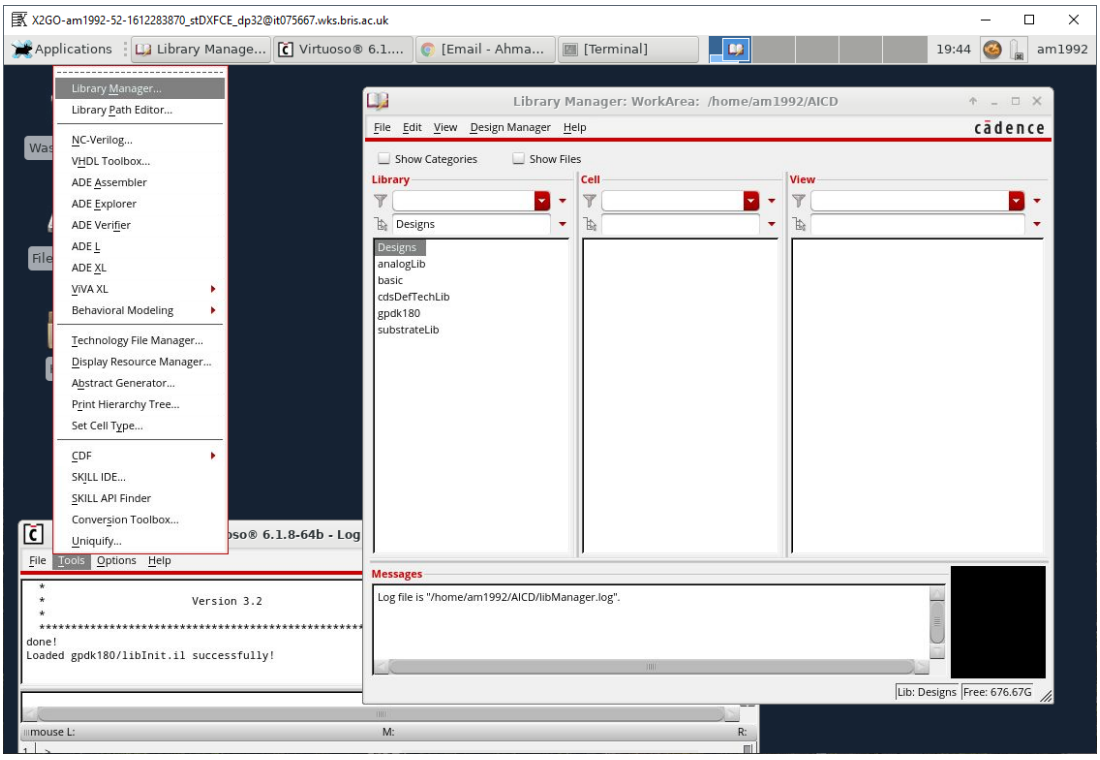


< Figure 4: The Command Interpreter Window (CIW).
The text displayed in your CIW may look different. The CIW starts up displaying only a few lines of output. You can resize the window or use the scroll bars to view more lines.>

(3.3.3) You can explore more of the CIW from the online help. From the CIW, choose **Menu: Help → User Guide.**

3.4 The Library Manager

(3.4.1). Here, we will Investigate the library manager and the different libraries and their components (e.g., res, gnd, vcvs, noConn, ISO border). From the tools menu of the CIW, launch the Library Manager from **Menu: Tools → Library Manager.**



< Figure 5: The Library Manager window with predefined libraries for this course. >

(3.4.2) You can use the Library Manager to create, add, copy, delete, and organise libraries, cells, and cellviews in a design project. The most important aspect to understand here, is how the Library Manager is organised: Each library typically holds several cells, and each cell typically has several views. We will describe each as follows.

1. **Libraries**. A Cadence design project organises itself in terms of libraries. A library in a design project serves a similar function to a folder/directory of an Operating System. Each library contains a number of components called cells. The cells in a library are typically related by a common theme (though this not strictly necessary, it is good practice). For example, you may have a library of analogue cells (FETs, BJTs, Diodes, resistors, caps…) and another library of digital cells (logic-gates, flip-flops, MUXs, registers…). Or, you may have a library of cells by one vendor/foundry (AMS, UMC, TMSC…), and another library from a different vendor/foundry. Libraries may be created by the user or predefined/included by Cadence or a third-party vendor.

2. **Cells**. Each cell in a library represents a component or part. These components may represent a passive or active primitive device (resistor, capacitor, diode, transistor…) or a system of devices (amplifiers, oscillators, regulators, gates, ALUs…). A cell may itself be composed of or reference other cells from the same or a different library. Some cells serve only the purposes of the schematic editor or simulation application (noConn, iprobe, ISO border…). These kinds of cells do not represent real devices or circuits—they are there only to facilitate the schematic entry design flow.

3. **Views**. Each cell has at least one view associated with it. Each view represents one way of modelling the component. For example, a cell may have a schematic view, which describes its logical connections, or a layout view, which describes its physical on-chip layout. A cell may also have symbol view, which is what you would use to instantiate that part in another schematic. A cell may also have a simulator view (functional or behavioural) used by the SPICE-based simulator.

(3.4.3) For this course, when you open the Library Manager you will notice six predefined libraries: **Designs**, **analogLib**, **basic**, **gpdk180**, **cdsDefTechLib**, and **substrateLib**. In practice, you would be required to create your personal design directory(s) and link to them, along with any vendor parts libraries you require, via the cds.lib library definition file in your AICD folder. However, this has already been done by the EENGM030_setup script you ran earlier. You therefore do not need to create or define any libraries.

(3.4.4) The predefined libraries for this course (that you should see in your Library manager) are explained below:

1. **Designs.** This library is where you will store all of your design work (cells/views) for this course. The Designs library is a user-defined library and can be found as a folder in you AICD directory (e.g., ~/AICD/Designs). It was created and linked-to in the cds.lib file by the EENG6030_setup script the first time you setup the Cadence environment.

2. **analogLib**. This is a Cadence provided general parts library. The cells in this library are ideal versions of the most common analogue components (e.g., resistors, capacitors,

sources, ground). They are behavioural/mathematical models. They are simulatable but not synthesizable or realizable and do not have a layout (physical) view. Each cell in this library has at least a symbol view (so that it can be instantiated for use in your schematic) and a **spectre** view (Cadence's SPICE-class simulator is called **Spectre**) that defines its simulation behaviour. (Some components have additional views for other simulators – **auCdl**, **auLvs**, **hspiceD**, **hspiceS** – which are not used in this course).

3. **basic**. This is another Cadence provided general parts library. It contains schematic and symbol views of various connectors, pins, and supplies. These are also ideal parts and are not synthesizable or realisable. Each cell in the basic library has at least a symbol view. Additionally, some cells may have a schematic and/or one or more simulator views (spectre, auCdl, auLvs, hspiceD, hspiceS) that defines its simulation behaviour. (You may also come across two other Cadence provided libraries – **ahdLib** and **rfLib** – but they are not used in this course and have therefore not been defined in the Library Manager).

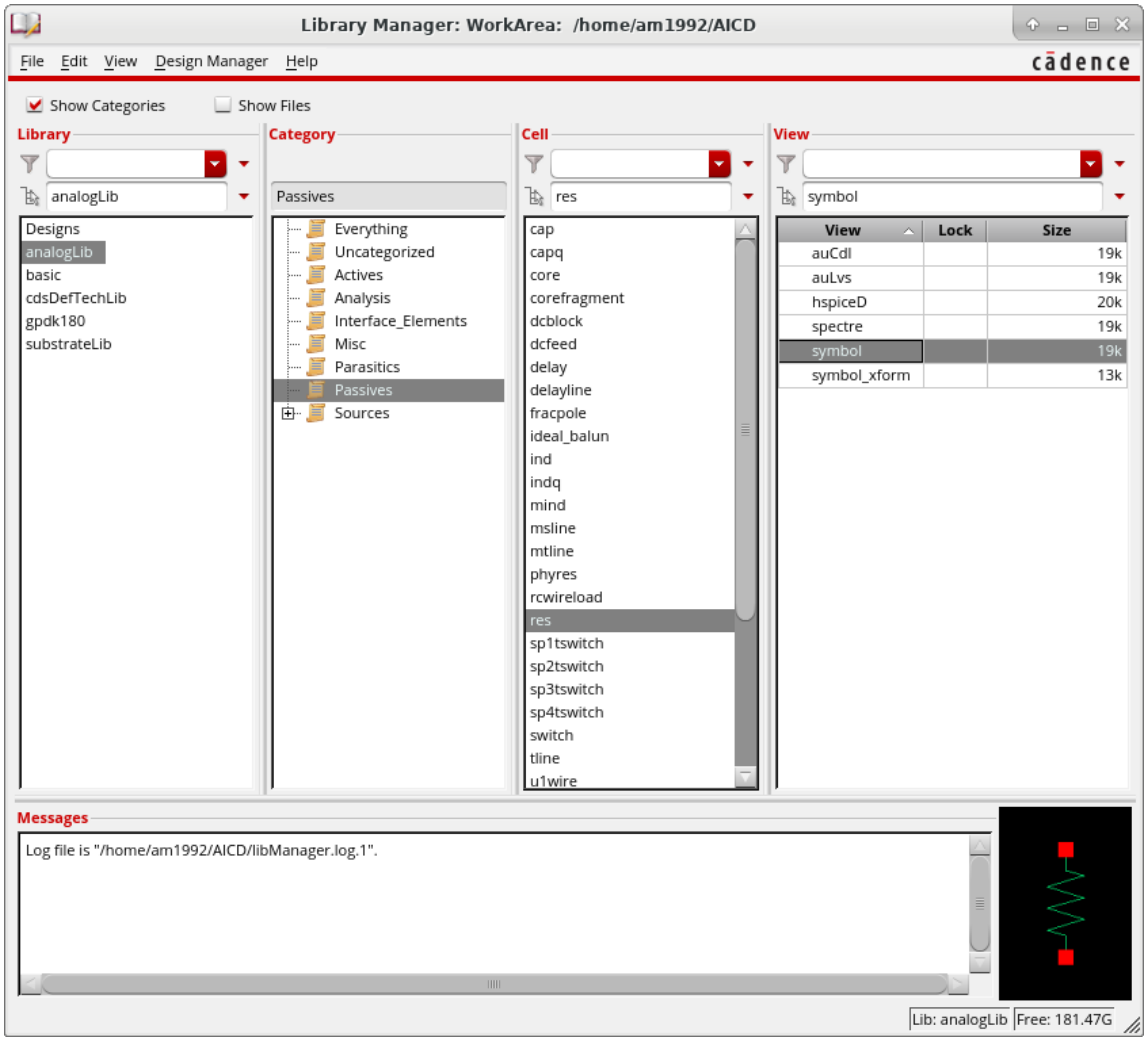
4. **gpdk180.** This is a process design kit (PDK) technology library. It contains schematic, symbol, layout, simulation models, and technology files of a vendor or foundry's 180 nm CMOS process technology. The GPDK180 is a generic PDK provided by Cadence— it is not a real process technology produced by any real foundry and therefore is not synthesisable or realisable. It emulates the behaviour of a typical (generic) 180 nm MOSFET process node. The technology files are useful if you want to do layout.

5. **cdsDefTechLib**. This is the Cadence default technology database. It is referenced by any technology library in your design project, in this case by the GPDK180 library. The details of the library are beyond the scope of this course.

6. **substrateLib.** This library is associated with the GPDK180 technology library. It contains cells which are used during substrate extraction process for parasitic and/or noise behaviour. The details of the library are beyond the scope of this course.

(3.4.5) Notice at the top of the Library Manager there are two check boxes 'Show Categories' and 'Show Files'. The Show Categories check box organises the contents of each library into specific categories. Each category can be further divided into subcategories. This can make it easier to find the component you're looking for. It is often easier to just text search for the component. Each section of the Library Manager (Library, Category, Cell, and View) has a text search field just above the listing area. The 'Show files' check box can be ignored (it lists all the internal files associated with a particular library, category, or cell).

(3.4.6) Explore the '**analogLib**' ideal parts library. In this course, you will make heavy use of **res** (passives: resistor), **cap** (passives: capacitor), **vdc** (sources: independent: DC voltage source), **idc** (sources: independent: DC current source), **gnd** (sources: globals: ground). We will also make use of the controlled source **vcvs** (sources: dependent: voltage-controlled-voltage-source). Note that the analogLib library is divided into nine categories when the 'Show Categories' box is checked (Everything, Uncategorized, Actives, Analysis, Interface_Elements, Misc, Parasitics, Passives, and Sources). The 'Sources' category is further divided into five sub-categories (Dependent, Globals, Independent, Ports, Z_S_Domain).



< Figure 6: Library Manager with 'Show Categories' check box selected. Note: when you select a schematic or symbol cellview an illustration of the cell shows in the bottom-right corner of the dialog box. This doesn't always work as it should! >

(3.4.7) Also explore the '**basic**' and '**GPDK180**' libraries. These two libraries, like analogLib, are also categorised. In these labs, we will only be using the **nmoshv** n-type MOSFET model and the **pmoshv** p-type MOSFET from the gpdk180 library. Note, that these are four-terminal devices (gate, source, drain, and body/bulk). Note the 'hv' suffix stands for high voltage. This is relative! For our purposes, high voltage means a maximum VDS of 2.5V. This is in contrast to low-voltage devices with a sub-1V VDS. (The high voltage behaviour is achieved by using a thicker gate oxide level.) Note the symbol, spectre and layout views in the gpdk180.

(3.4.8) From the '**basic**' library you will only be making use of the **noConn** (no-connection) part. Note that while the analogLib/gpdk180 parts have a layout/spectre view (layout and simulation behaviour), the noConn part does not have a layout or spectre view as it only used by the schematic editor to avoid warning messages on floating wires (it does not simulate). The noConn part will be demonstrated later in this lab.

(3.4.9) You can explore more of the Library Manager from the online help. From the Library Manager, choose **Menu: Help → User Guide.**

4. Beginning the Circuit Simulation Exercises

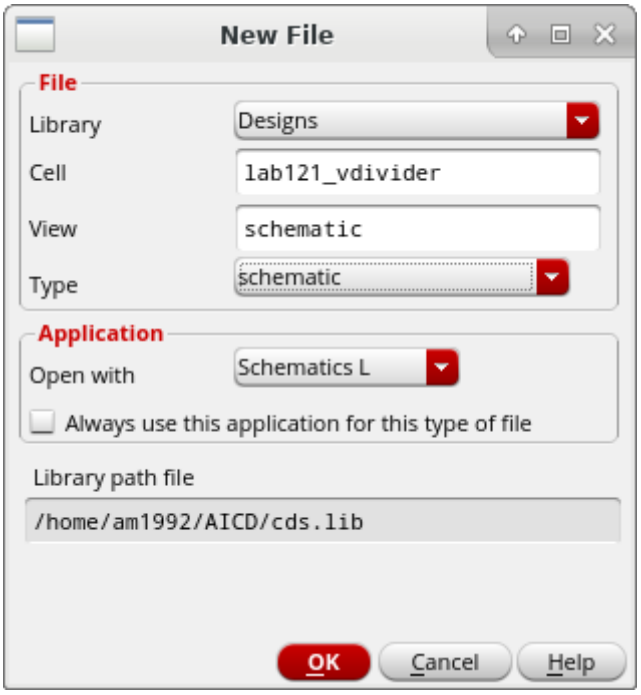
In this section, we will

- Introduce the Virtuoso Schematic Editor and the Analogue Design Environment (ADE) from the Cadence toolset.

- Cover the basic steps of the schematic entry-driven, SPICE-based IC design flow by creating and simulating a simple voltage resistive divider circuit with an ideal DC voltage source supply.

4.1 Creating a New Schematic

(4.1.1) From the Library Manager window, select the 'Designs' library and choose **Menu: Files: à New à Cell View**. You should see the 'New File' dialog box (fig. 6, this box can sometimes hide itself behind the Library Manager window – so check if it doesn't seem to appear).



< Figure 7. The New File Dialog Box >

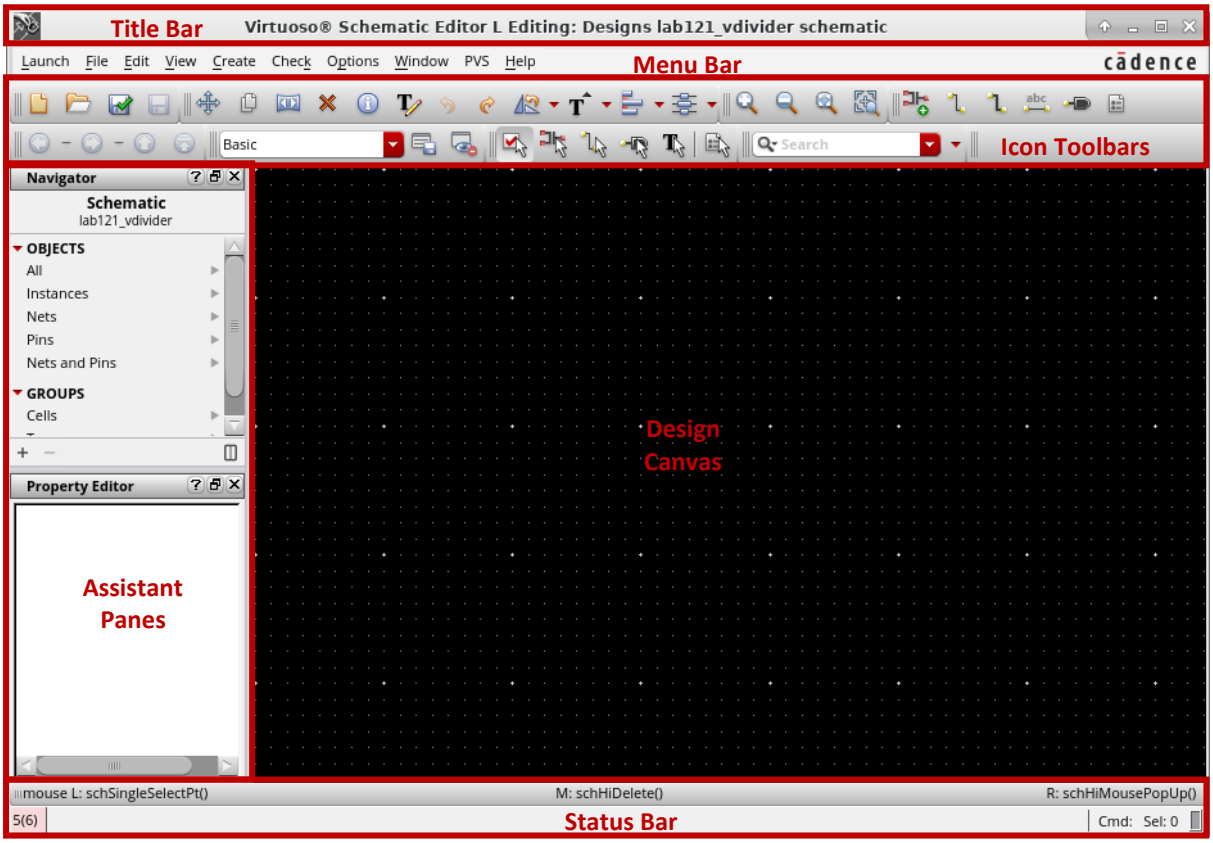
(4.1.2) In the 'Cell' text field, choose a meaningful cell name (e.g. lab121_vdivider). In the 'View' text field, type 'schematic' as the view name and from the 'Type' drop down menu choose 'schematic' as the view type. In the 'Application' section of the dialog box choose 'Schematics L' from the 'Open with' drop down menu. OK the form.

(4.1.3) You should now be presented with the **Virtuoso Schematic Editor L** schematic entry window for the cell you just created (fig. 8). If you check the Library Manager, you will see that a cell called 'lab121_vdivider' (or whatever name you chose) with view name 'schematic' has been added to the 'Designs' library. This cell view will not be saved in the Library Manager

until you check and save it in the Schematic Editor. From the Schematic Editor window, choose **Menu: File à Check and Save**. (Or, you can also **Menu: Check à Current_Cellview** followed by **Menu: File à Save** to perform the same command.)

(4.1.4) Note the many options available from the file ('type') and application ('open with') drop down menus in the new file dialog box. These are how you can add new cellviews to your pre-existing libraries, and also add new views to your pre-existing cells. With experience, you will learn that there are many different ways of adding cellviews to your libraries.

(4.1.5) Also note that some of the tools of the Cadence environment (see **Menu: Launch**) are divided into three tiers L, XL, and GXL e.g. (Schematic Editor L and Schematic Editor XL). Successive tiers (L → XL → GXL) are versions of the same tool, with additional/advanced features. You will only use Schematic Editor L and ADE L tools in this course. The L/XL/GXL don't stand for anything.



< Figure 8. The Virtuoso Schematic Editor L Window.
Note the components of the editor – Title Bar, Menu Bar, Tool Bars, Assistant Panes, Design Canvas, and Status Bar. >

PLEASE NOTE: IF A SCHEMATIC ISN'T PROPERLY CLOSED DOWN (FOR EXAMPLE, DUE TO A DROPPED REMOTE CONNECTION), THE SCHEMATIC MIGHT BECOME LOCKED. IF THIS HAPPENS YOU WILL NOT BE ABLE TO OPEN THE FILE FOR EDITING. YOU WILL ONLY BE ABLE TO OPEN THE FILE FOR READING UNTIL YOU REMOVE ANY/ALL LOCK FILES. TO DO THIS, GO TO YOUR '~/AICD/Designs' FOLDER AND FIND THE FOLDER REPRESENTING THE SCHEMATIC E.G. 'lab121_vdivider'. DELETE ALL THE .cdslck LOCK FILES AND YOU SHOULD BE ABLE TO OPEN YOUR SCHEMATIC FOR EDITING.

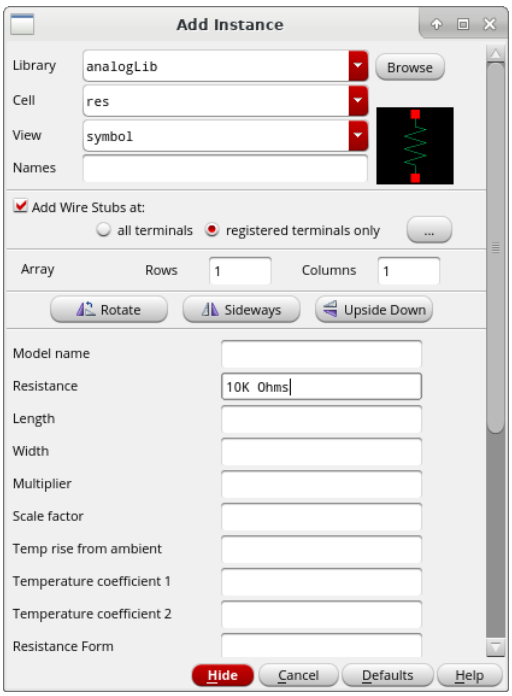
4.2 Adding Components to the Schematic

(4.2.1) On the design canvas of the Schematic Editor, we will place two 10KΩ resistors (**res**), one 10V DC voltage source (**vdc**), and a ground connection (**gnd**). We will then wire them to create a resistive voltage divider circuit. All parts are sourced from the **analogLib** library.

(4.2.2) From the Schematic Editor window, choose **Menu: Create → Instance...** This will open the '**Add instance**' form (fig. 9). Note: There are multiple ways of starting a command in the schematic editor. For example, to add a new part you can use the menu bar **Menu: Create → Instance...**, or you can use the '**Create Instance**' icon-button on the toolbar at the top of the schematic window (fig. 11), or you can use the keyboard shortcut '**I**' (the shortcut key – if one exists – will be shown next to the menu item). You can also right-click anywhere in the empty space of the design canvas to invoke a context-menu and select the **Add Instance…** command from there. Becoming familiar with all available methods will improve your productivity!

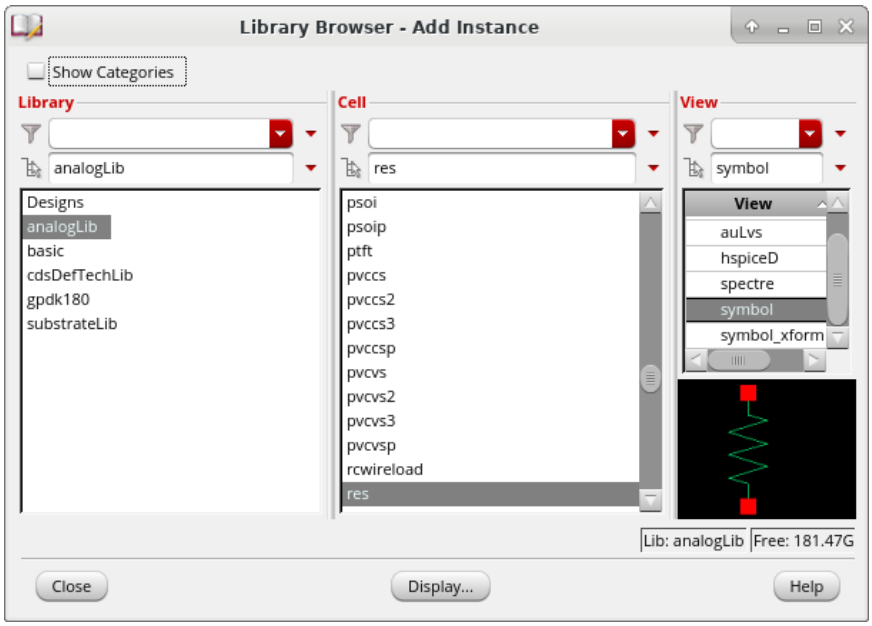
(4.2.3) In the 'Library' text field type 'analogLib', in the 'Cell' text field type 'res', and in the 'View' text field type 'symbol'. (The fields may autocomplete as you type). When you have filled in these text fields, the **Add Instance** form will expand listing the attributes/parameters (and their text fields) for the cell, in this case **res**. (You may need to click away from the current text field for the form to expand.)

(4.2.4) You will see that default values are filled-in for some fields on the **Add Instance** form, others are empty. In the case of the resistor (**res**), the only default value is for the 'Resistance' attribute (1K Ohms). Change this to 10K Ohms. **Do NOT enter the unit type 'Ohms' into the text field. This is automatically appended by the software when you click away from the text field. For a 10KΩ resistor, just enter 10k, 10K, or 10000 in the Resistance text field.** Explore the other fields. Leave them empty. Click the 'Hide' button to exit the form.

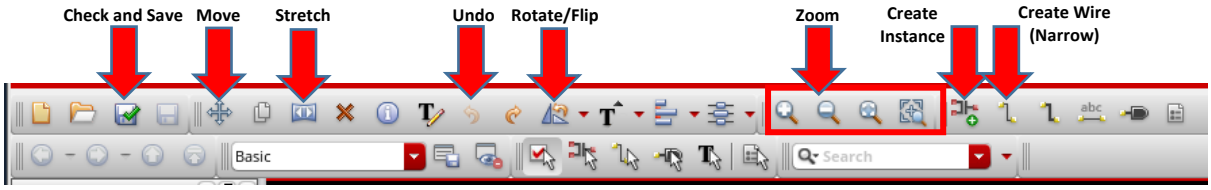


< Figure 9. The Add Instance Form. Filled in for a 10KΩ resistor.>

(4.2.5) Note: You can also use the browse button to select the required library, cell, and view. Click 'Browse'. The 'Library Browser - Add Instance' window will open. Select the **analogLib** library, **res** cell, and **symbol** view. See Figure 10. You can check the 'Show Categories' box to make the search easier or you can text search for the part using the text field in the Library, Cell, and View sections just above the listing area. After selecting the **res** cellview close the Library Browser and you will be returned to the Add Instance form.



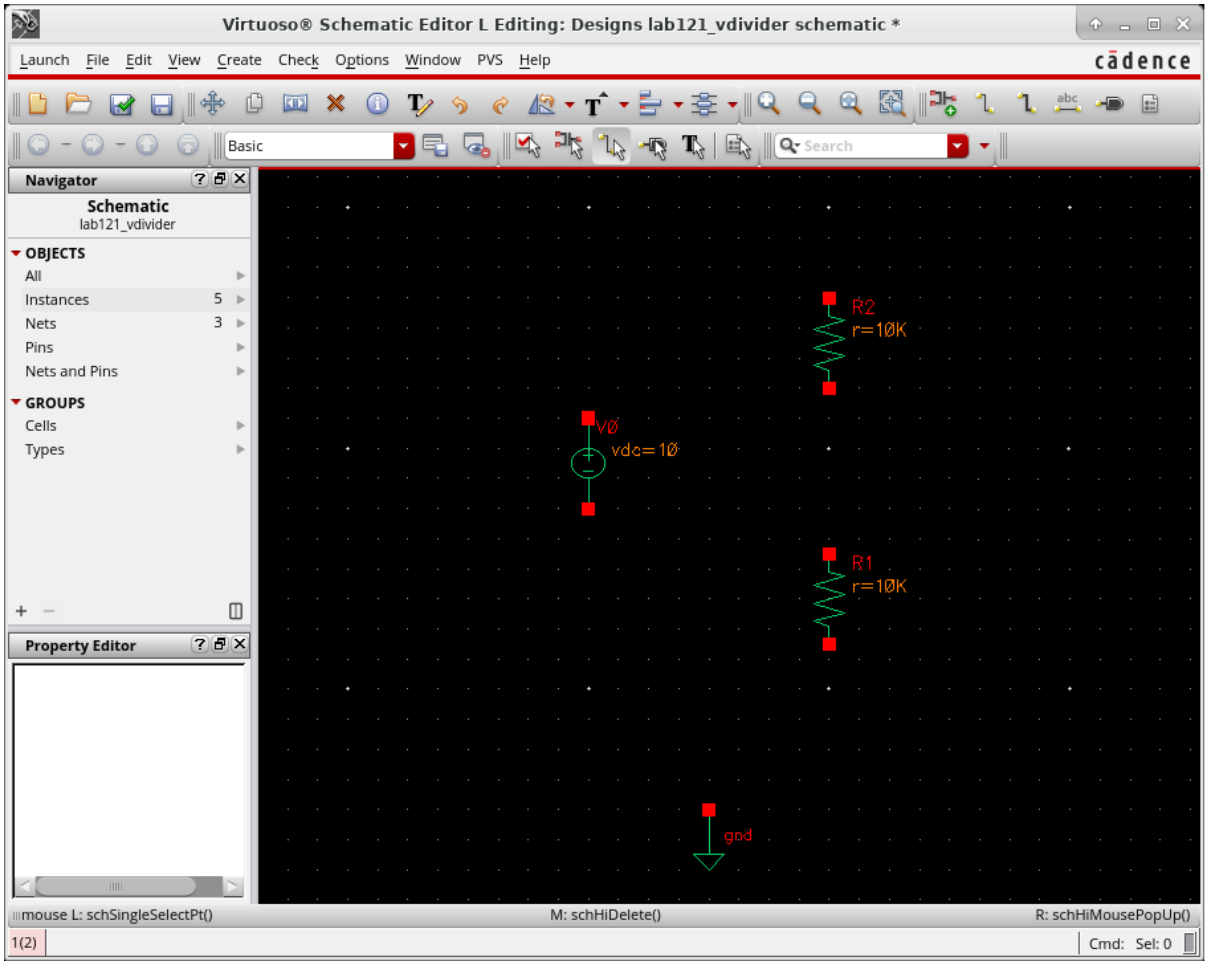
< Figure 10. The Library Browser. Note: that while the Library Browser looks similar to the Library Manager, they are not the same. You cannot use the Library Manager to place components in your schematic. Do not fall into the trap of opening an analogLib cellview from the Library Manager! >



< Figure 11. Useful icon-buttons. Note: to see what an icon button does, hover over the icon and the name of the button will appear. >

(4.2.6) On hiding the **Add Instance** form, you will see a yellow resistor symbol moving with the mouse pointer. You can click anywhere on the design canvas to place as many instances of this component as required (in our case, two). Once placed onto the canvas, the colour of the resistor will change from yellow to green/red. **To stop placing instances of the resistor component, press the 'ESC' key.**

(4.2.7) Now add the remaining two components from the analogLib library – **vdc** and **gnd** – using the same procedure as above. For the **vdc**, set the DC voltage text field to 10 (do not add the unit type), leave all other text fields unchanged (empty). Note the **gnd** component has no text fields.



< Figure 12. The analogLib components of the voltage divider circuit in the schematic editor. >

(4.2.8) Note: you can **zoom-in** and **zoom-out** of the design by using the scroll wheel. You can **pan** the design canvas in any direction by holding down the middle button (scroll wheel) and dragging the canvas in any direction. To centre the view (fit-to-screen), you can press **F**. There are a number of zoom options – zoom-in, zoom-out, zoom-selected, fit-to-screen– accessible from the 'View' menu (**Menu: View → …**) and also the icon-button toolbar.

(4.2.9) Note: You can undo a command by selecting **Menu: Edit → Undo**, using the '**Undo**' icon-button on the toolbar, or by using the keyboard shortcut '**U**'. Your command history only exists between schematic saves/checks. You can undo all commands in your command history until you save or check the schematic (i.e., Save, Check, Check and Save). Once you have saved/checked your schematic your command history is cleared, and you can no longer undo any changes you may have made to your schematic.

(4.2.10) Note: When adding an instance to your schematic, it is good practice to place them in their (approximate) intended location at instantiation. If you want to move a component after placing it on the schematic, you can select it (left-click) and drag it to a new location. Alternatively, you can move the item by selecting **Menu: Edit → Move**, using the '**Move**' icon button on the toolbar, or the keyboard shortcut '**Shift+M**' and clicking on the item. You can also use the stretch command by selecting **Menu: Edit → Stretch**, the '**Stretch**' icon-button

on the toolbar, or the keyboard shortcut '**M**' and clicking on the item. The difference between **Move** (**Shift+M**) and **Stretch** (**M**) is that 'Move' will disconnect the part from its connected wires when moved, whereas 'Stretch' (like dragging) will move the wires with it.

(4.2.11) Note: When adding an instance to your schematic you can rotate and/or flip the item before you place it on the canvas using the 'Rotate', 'Sideways', and 'Upside Down' buttons on the Add Instance form (refer back to Figure 9). Alternatively, you can rotate or flip the item before placing it on the canvas using the keyboard shortcut '**R**'.

After placing a component on the canvas, you can rotate the item using **Menu: Edit → Rotate** or you can use the **Rotate-Left/Rotate-Right/Flip-Horizontal/Flip-Vertical** multi-function icon-button on the toolbar at the top of the schematic window. Note that the functionality of the **Menu: Edit → Rotate** command and associated keyboard shortcut '**R**', is actually set by the Rotate/Flip multi-function icon-button – so whether this command Rotates-Left, Rotates-Right, Flips-Horizontal, or Flips-Vertical depends on the selection of this icon-button.

Finally, pressing F3 while a command is in progress gives you a more detailed set of options for that command. When moving an item using the **Move** or **Stretch** commands, pressing F3 will bring up a dialog box with more detailed options including 'Rotate', 'Sideways', and 'Upside Down' buttons. This is another way of rotating/flipping a component after placement.

(4.2.12) Note: You can modify the attributes of a component (e.g., Resistance or DC Voltage) after it has been placed on the design canvas, by first selecting the item (left-click on the item, you will see a pink box around the item when it is selected) and then by right-clicking on the item and selecting 'Properties…' You can also use **Menu: Edit → Properties → Objects**, the '**Edit Properties**' icon-button from the toolbar, or the keyboard shortcut '**Q**'. You can then modify the required attribute(s) and Apply/OK the change.

4.3 Nets, Nodes, and Wires.

(4.3.1) With the components placed (as in Figure 12), you can now add wires to connect them together. You can use **Menu: Create → Wire (narrow)**, the '**Create Narrow Wire**' icon-button on the toolbar, or the keyboard shortcut '**W**' to start the wire command.

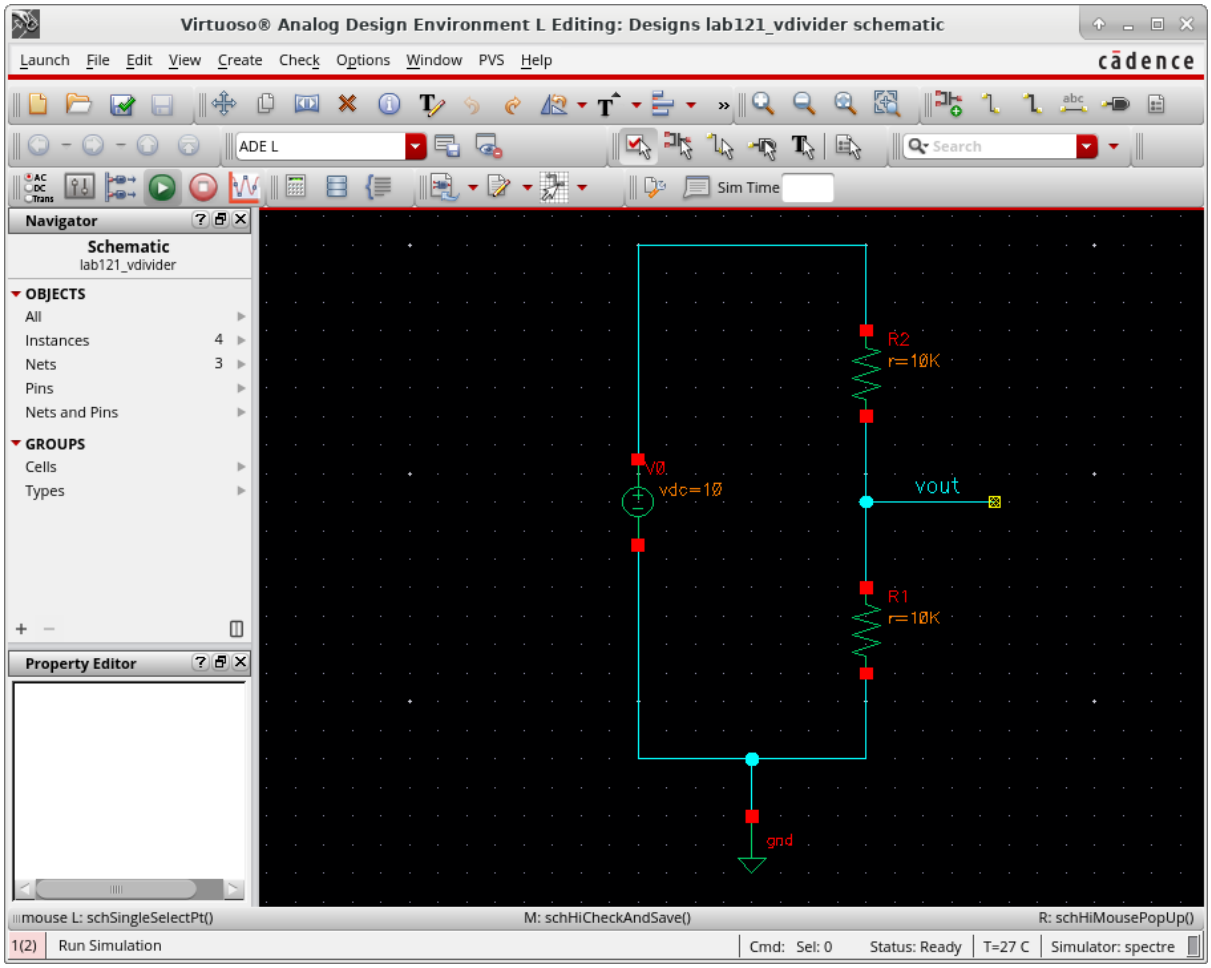
(4.3.2) A wire connects two or more **nodes** and/or **nets** together to form a single **net**. In SPICE terminology, a **net** is the interconnection between one or more components. A **net** has the same potential (voltage) with respect to ground at all points along the interconnection. A **node** identifies a terminal of a component into which a current flows. In the schematic editor, a **net** is created by the wires (blue lines) connecting one or more components. A **node** is identified by the small red squares at the terminals of a component.

(4.3.3) To wire two components together, left-click once on a terminal node of the first component (the small red squares) to begin the wire and then left-click again on a terminal node of the second component to terminate the wire. **For each component click once on the node and release, do not drag!** The nodes should now be connected together with a wire.

(4.3.4) Repeat this for all connections of the divider circuit (see Figure 13). You can also wire a component by placing your cursor on the terminal node of a component (the small red squares) and dragging away from the component while holding down the left-mouse button – begin at the terminal node of one component and terminate on the terminal node of the second component.

(4.3.5) Note: Every net is identified by a net name. By default, this is 'net' appended by a number. The net names are sequentially generated e.g., net1, net2, net3… Some nets have special names e.g., ground (gnd!). The net names are hidden by default – you can toggle on/off the visibility of net names from **Menu: View à Annotations à Net Names**. Or right-click on the canvas to invoke the context-menu and select **Annotations à Net Names**.

(4.3.6) In most cases you can change the default net name to something more helpful. Extend the output net (the net between the two resistors) to the right and name it something intuitive for graphical output (see Figure. 13).



< Figure 13. The wired voltage divider circuit in the schematic editor. >

(4.3.7) Choose **Menu: Create à Wire Name**, use the '**Create Wire Name**' icon-button, or the keyboard shortcut '**L**' to start the create wire name command. The 'Create Wire Name' form will appear, type 'vout' in the 'Names' text field and 'Hide' the form. On hiding the form, you

will see the name ‘vout’ in yellow moving with the mouse pointer. Click on the output net to assign the new name to it. The name will change from yellow to blue (the colour of the wire) when it has been successfully associated with the wire.



< Figure 14. The Create Wire Name Form. >

4.4 Checking and Saving the Schematic

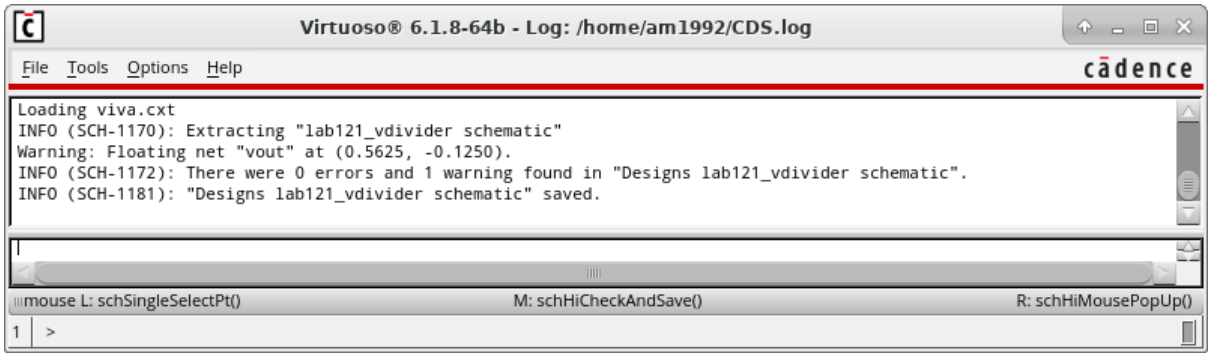
(4.4.1) Save your schematic. You can use the command **Menu: File → Check and Save**, the ‘**Check and Save**’ icon-button, or the keyboard shortcut ‘**Shift+X**’.

(4.4.2) Note: When you check and save you will see a warning message. You will also notice that the unconnected end of the output net (vout), has been highlighted with a yellow square. The schematic editor will issue warnings if there any issues with your circuit. Warnings are not fatal. Your circuit will still be saved and can still be netlisted and simulated. Warnings identify some ambiguity in your circuit such as unconnected pins, wire crossovers, overlapping components and many more. It is good practice to address and eliminate all warnings. Errors are fatal. A schematic cannot be netlisted or simulated if errors exist.

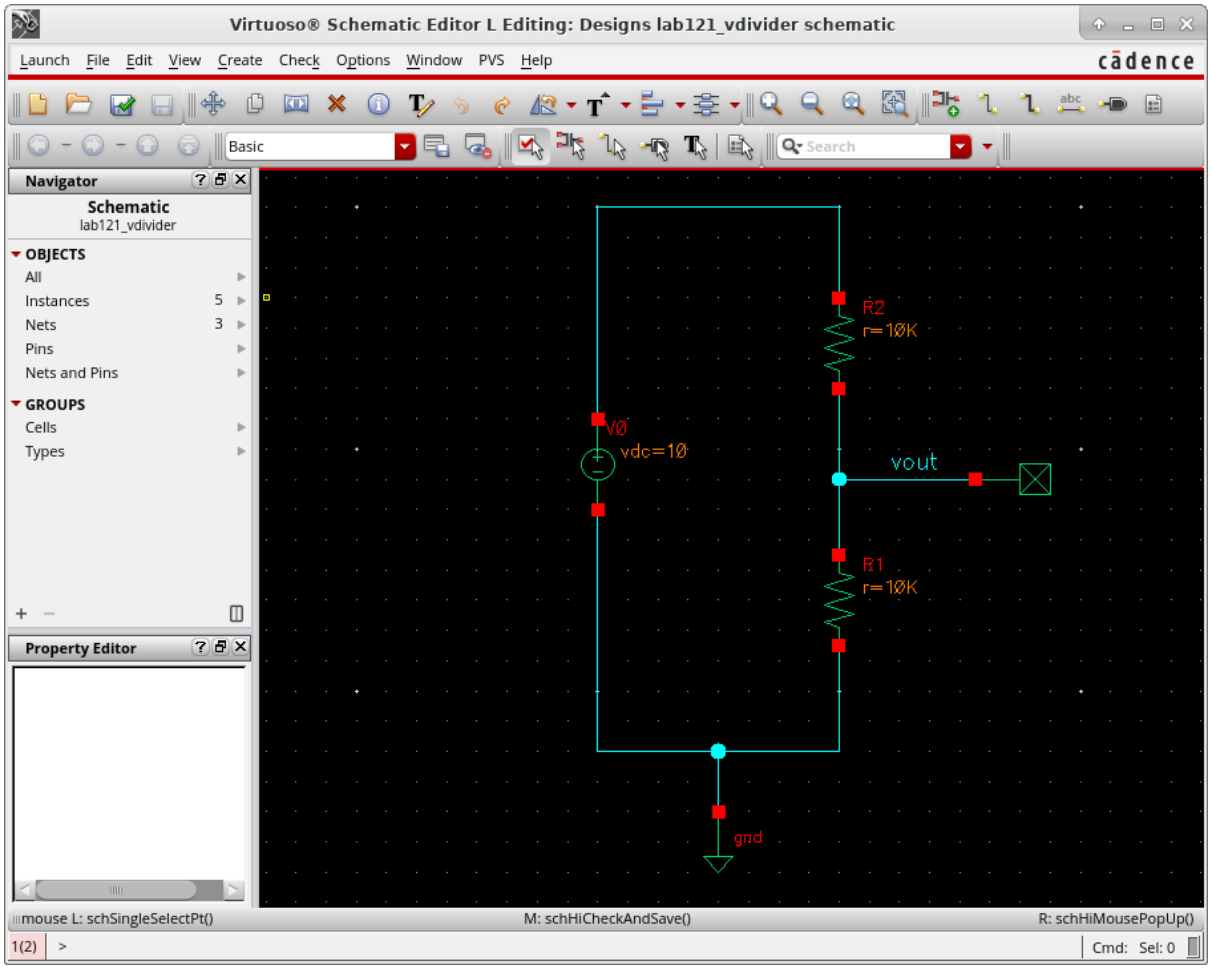
(4.4.3) Note: If there are any warnings or errors generated by your schematic, they will appear in the output area of the CIW. Always check the CIW to diagnose any errors or warnings.

(4.4.4) View the output area of the CIW. Note the ‘floating pin’ warning line (Figure 15.). To avoid floating net warnings, add the **noConn** part from the basic library (Library: **basic**, Cell: **noConn**, View: **symbol**) at the end of any unconnected nets and/or nodes. See Figure 16.

(4.4.5) **Check and Save** your schematic again. This time there should be no warning.



< Figure 15. 'Floating net' warning after Check and Save. >



< Figure 16. The complete voltage divider circuit in the schematic editor including noConn (no-connection) to avoid 'floating net' warnings. >

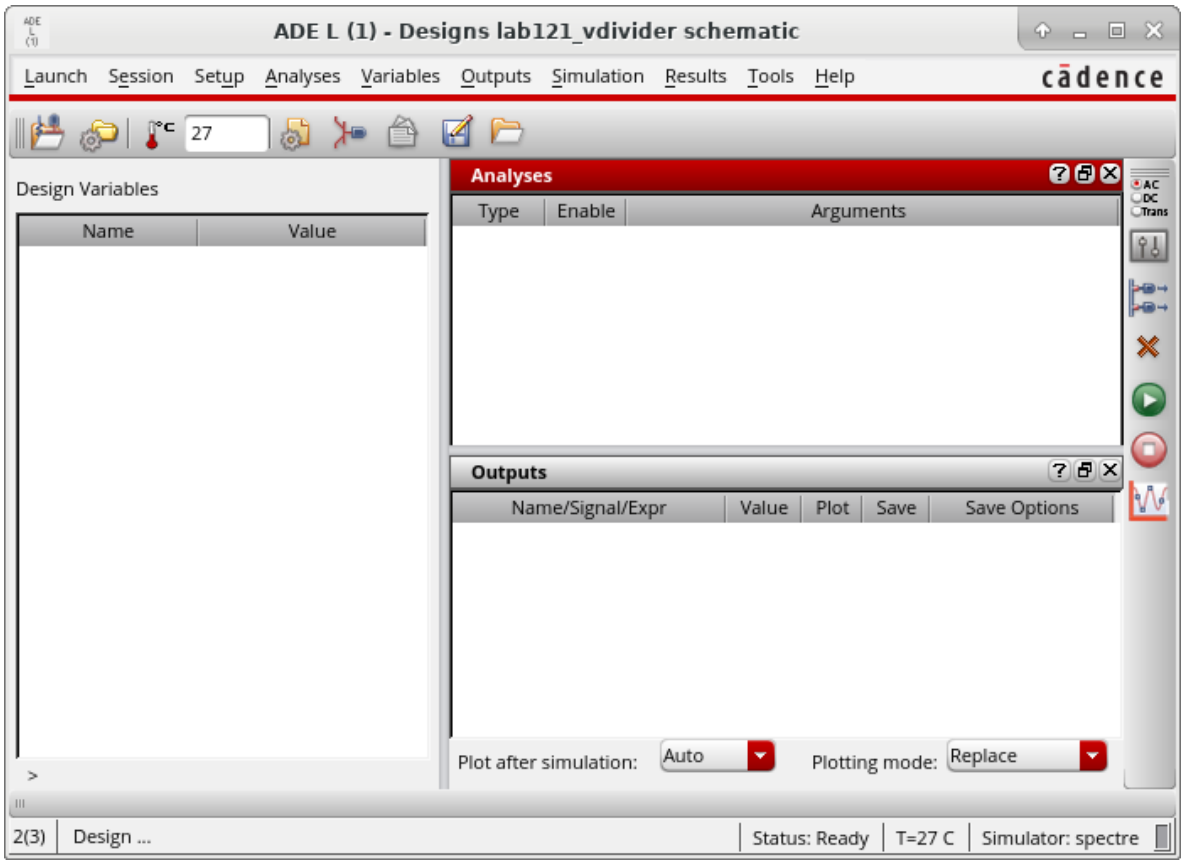
(4.4.6) Note: There is another save option **Menu: File → Save**. The difference between **Save** and **Check and Save**, is that the latter will check the schematic for correctness (errors or other issues) before it saves. If there are errors, it will ask you to confirm that you still wish to save. **Save** does not check your schematic for any issues – it will save your schematic regardless of correctness.

(4.4.7) Note: A netlist is required to simulate your design. A netlist is a text-based description of the connections of your circuit generated from your schematic. The Cadence netlisters will not generate a netlist until the schematic has been checked – if you do not check your design after making changes, it will not simulate. To only check your design, you can use **Menu: Check → Current Cellview** (or keyboard shortcut **X**). If there are any warnings or errors generated by your schematic, they will appear in the output area of the CIW.

4.5 Simulating the Schematic

(4.5.1) You should now have the completed schematic of the voltage divider circuit as shown in Figure 15. It should check and save without any errors or warnings.

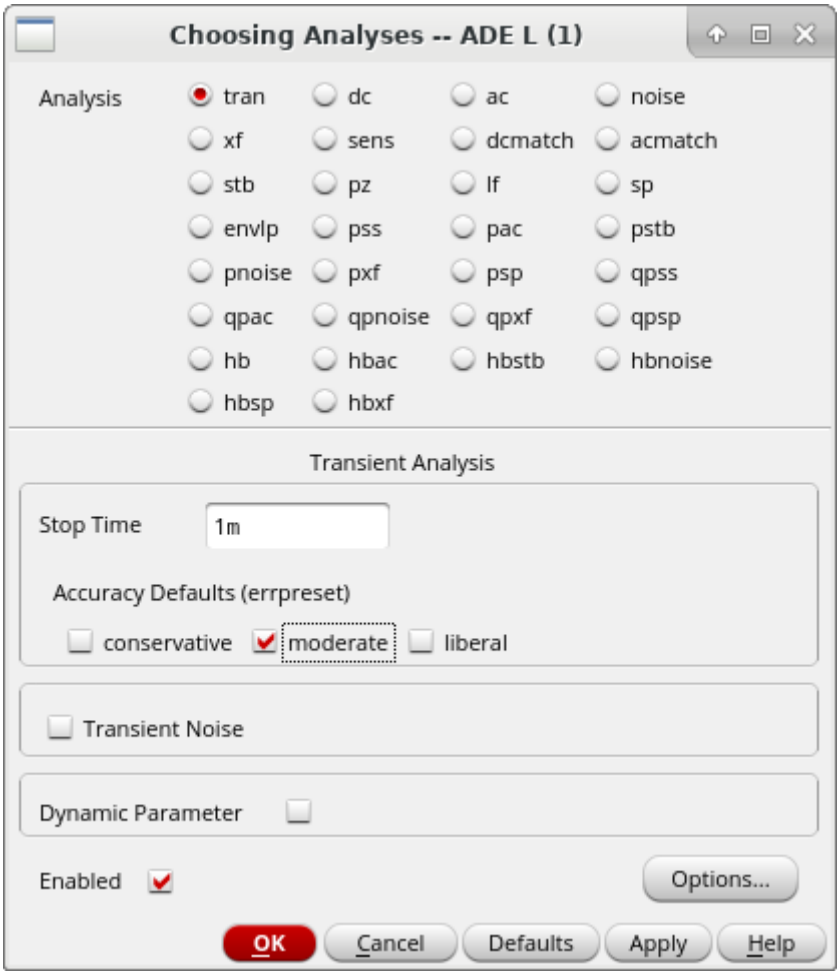
(4.5.2) We will now simulate the voltage divider circuit. From the Schematic Editor window, launch the Virtuoso Analogue Design Environment (ADE) window from **Menu: Launch → ADE L**. When you launch the ADE from a Schematic Editor window, the ADE will be pre-initialised for the design in that window, in this case the voltage divider circuit. This will be reflected in the title bar of the ADE window. (Alternatively, the ADE may be launched from the CIW, by **Menu: Tools → ADE L**. If you launch the ADE from the CIW you must manually setup the ADE for that design by selecting **Menu: Setup → Design…** and selecting the library, cell, and view).



< Figure 17. The Analogue Design Environment L (ADE L) Window. >

(4.5.3) Note: The ADE is the simulation environment for the Virtuoso toolset. The ADE is used to control all aspects of a schematic's simulation; including netlisting, starting and stopping a simulation, the types of analyses to run, which output waveforms (nets and nodes) to save and/or plot, the range of component values over which to simulate, circuit temperature and temperature variations, technology process variations, and much more. The ADE controls the settings for the **Spectre** simulator (Cadence's SPICE-class simulator).

(4.5.4) To simulate a design, we first need to choose an analysis type. From the ADE L window, go to **Menu: Analyses à Choose…** or use the '**Choose Analyses**' button on the right-side of the ADE window (top button). The 'Choosing Analyses – ADE L' form will open. Note: to see what an icon-button does, hover over the button and the name of the icon will appear.



< Figure 18. The Choosing Analyses Form. >

(4.5.5) Note: Many of the analyses options listed on the '**Choosing Analyses**' form are beyond the scope of this course. In this course, we will stick to the four fundamental SPICE analysis types – Transient Analysis (**tran**), DC Analysis (**dc**), AC Analysis (**ac**), and Stability Analysis (**stb**). From the Spectre user guide,

1. ***tran****. The Transient analysis computes the transient (time-domain) response of the circuit over a specified time interval.*

2. ***dc****. The DC Analysis finds the DC operating point or DC transfer curves of the circuit. To generate transfer curves, a parameter and a sweep range can be specified. The parameter can be a temperature, a design variable, a device instance parameter, or a device model parameter.*

3. ***ac****. The AC Analysis linearises the circuit about the DC operating point and computes the steady-state response of the circuit to a given small-signal sinusoidal stimulus. This analysis is useful for obtaining small-signal frequency response of a design.*

4. ***stb****. The Stability Analysis linearises the circuit about the DC operating point and computes loop gain, gain margin, and phase margin for a specific feedback loop or an active device. The stability of the circuit can be determined from the loop gain waveform.*

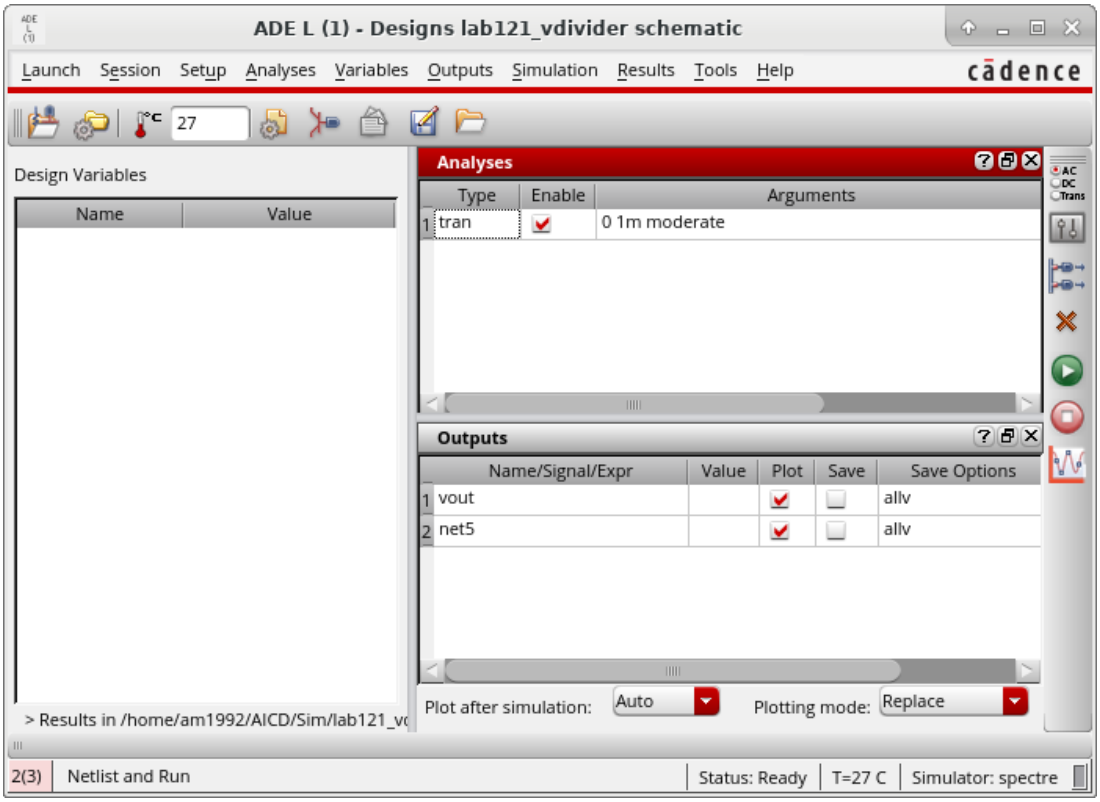
(4.5.6) We will start with a transient analysis. From the '**Choose Analyses**' form, select the '**tran**' option button and enter '1m' (one millisecond) in the 'Stop Time' text field. Set the 'Accuracy' option button to 'moderate'. Ensure that the analysis is enabled by checking the 'Enabled' box near the bottom of the form. See Figure 18. OK the form. And the **tran** analysis should appear in the 'Analyses' section of the ADE window.

(4.5.7) Note: The accuracy setting allows the designer to trade off the speed and accuracy of a transient analysis with a single setting: liberal, moderate, or conservative. The Liberal setting is fastest but has lowest accuracy. The conservative setting is the most accurate but also the slowest. Moderate is a compromise between liberal and conservative accuracy. From the spectre user guide…*The liberal setting is suitable for digital circuits or analog circuits that have only short time constants. The conservative setting is appropriate for sensitive analog circuits.* We will demonstrate the difference between these settings later in this lab.

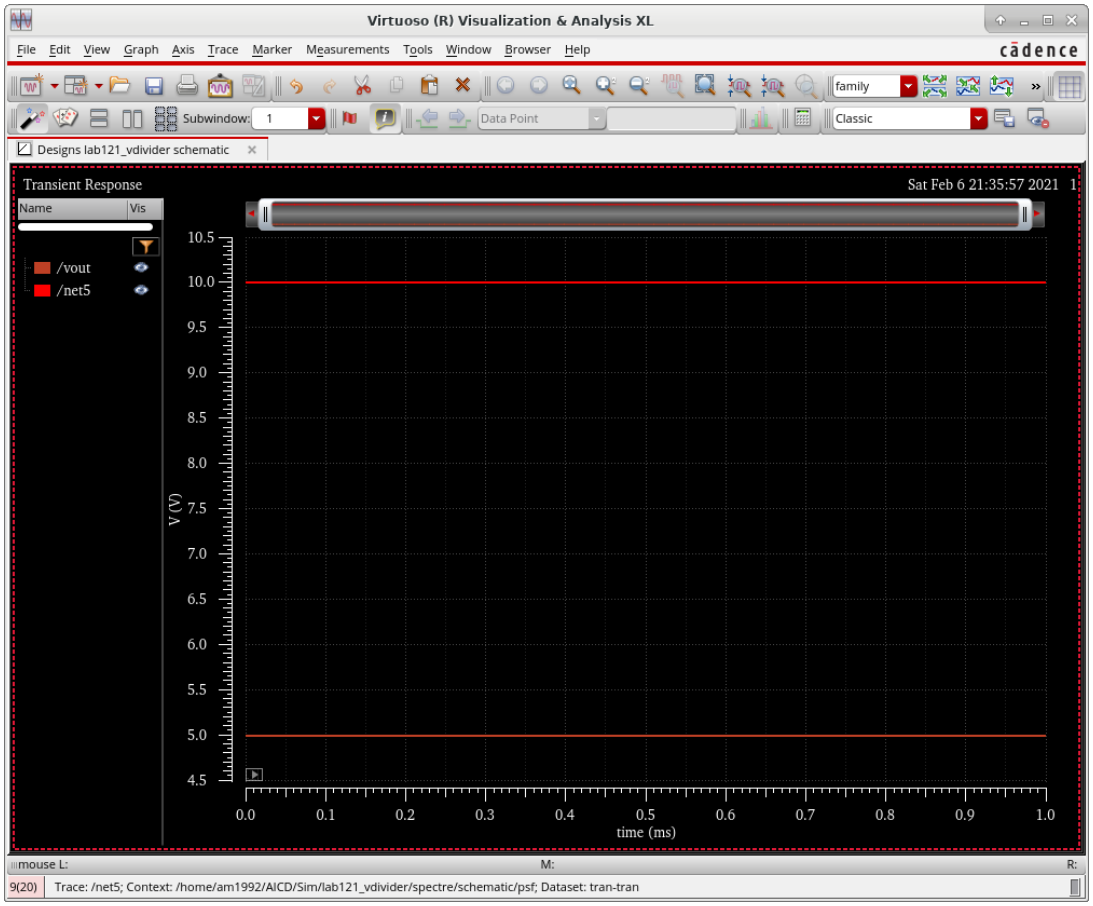
(4.5.8) We will now select the net voltages we want to plot. In the Analog Design Environment window, select **Menu: Outputs à To Be Plotted à Select On Schematic** to enter the plotting mode of the ADE**.** The schematic window will automatically become current. Select the voltage supply and output nets (blue wires) by clicking on them one after the other. The nets will change colour as you select them. **When you have selected the nets you want to plot, you must always press the 'ESC' key to exit the plotting mode of the ADE.** If you forget to escape the plotting mode, every time you click on the schematic window you will continue to (inadvertently) add more waveforms to your output window!

(4.5.9) Note: As you click on the nets (wires) to plot, observe that they appear in the 'Outputs' section of the ADE window. The name of the net is displayed, along with two checkboxes that determine whether the waveforms for those nets are to be saved and/or plotted. See Fig. 19.

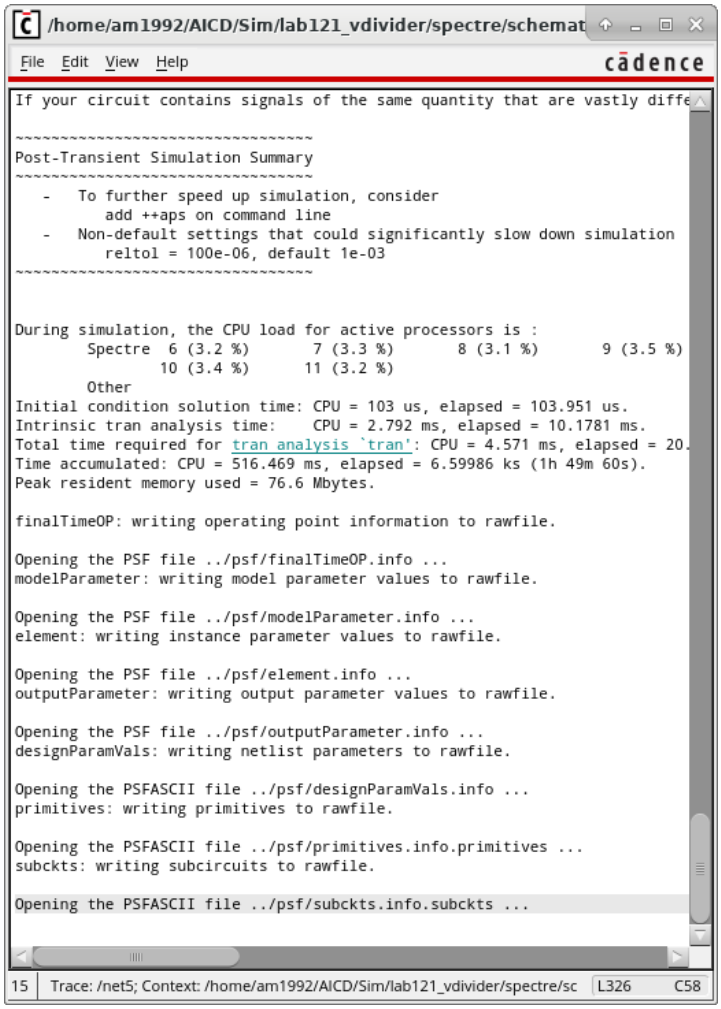
(4.5.10) In the Analog Design Environment window, run the '**Netlist and Run**' command by clicking on the green play button. If you followed all the instructions correctly, the simulation should run properly and both a textual output window as well as a waveform output window as shown in Figures 20 and 21 should open.



< Figure 19. The ADE with Analyses and Outputs set. >



< Figure 20. The Waveform Output Window. >



< Figure 21. The Textual Output Window. >

(4.5.9) Note: The operation of the resistive voltage divider is such that,

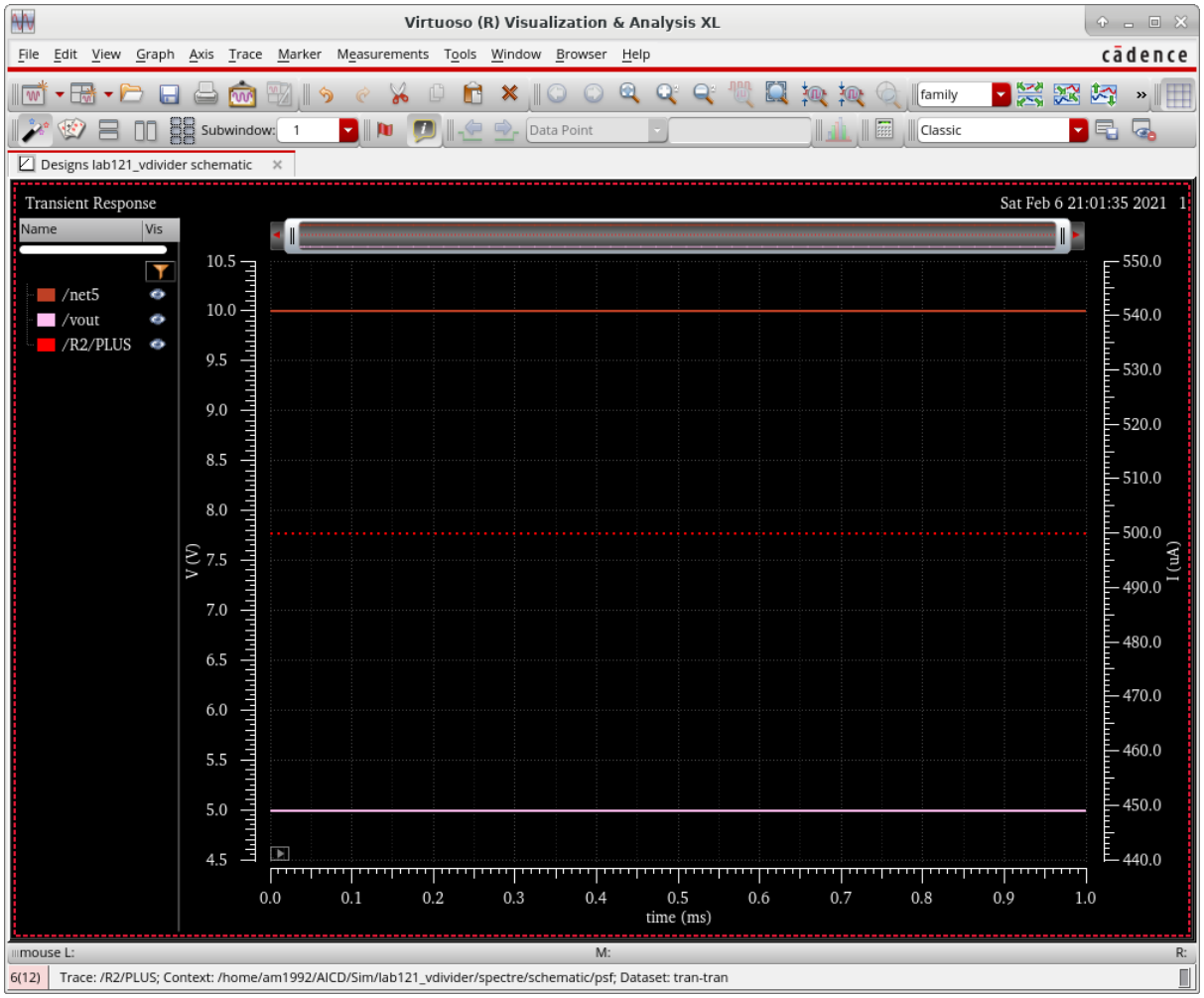
$$vout = vdc \times \frac{R1}{R1 + R2} = 5V$$

You can identify the supply (10V) and output (5V) waveforms by their net names on the left-side panel. Note that the output net is called 'vout' (or whatever name you chose) while the supply net name is the default net name 'netX' (where X is a number) since we did not name it. Since the voltage divider circuit uses only a DC voltage source, the waveforms do not vary with time.

(4.5.10) We will now add a branch current to the analysis output. Again, select **Menu: Outputs à To Be Plotted à Select On Schematic** to enter the plotting mode**.** If you want to save and plot the currents, you have to click a terminal **node** of a component (not the net/wire). Since, all components are in series, the current is the same at all nodes. Click on the terminal node of the upper most resistor (R2 in Figure 15 , yours may be different!). When you select a node, the node will turn pink. **Again, when you have selected all the nodes you wish to plot, you must always press the 'ESC' key to exit the plotting mode of the ADE.** The selected node should now be added to the 'Outputs' section of the ADE.

(4.5.11) Note: If in plotting mode, and you click on the body of a symbol, all terminal nodes of the component will be selected and added to your 'Outputs' window.

(4.5.12) A re-simulation is necessary, as by default, Cadence does not save currents. In the Analog Design Environment window, re-run the '**Netlist and Run**' command by clicking on the green play button. If the simulation runs properly, both a textual output window as well as a waveform output window will appear as shown in Figure 22 should open.



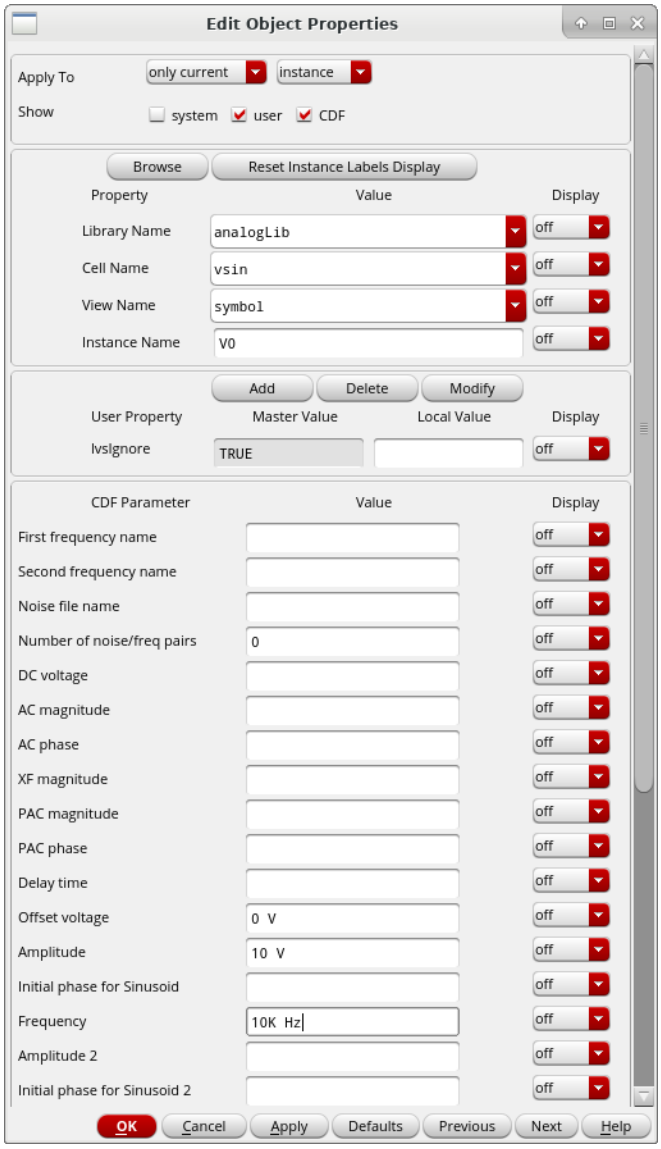
< Figure 22. The Waveform Output Window. >

(4.5.13) Note: The operation of the resistive voltage divider is such that,

$$i = \frac{vdc}{R1 + R2} = 0.5mA = 500\mu A$$

Identify the supply (10V), output (5V) waveforms, and node current (500uA) by their net/node names on the left-side panel. Note that the output node name by default is of the form '/component_name/terminal_name', in this case '/R2/PLUS' (i.e. the +ve terminal of resistor R2). Note that different waveform types/units (e.g. current uA, voltage V) will be shown on their own y-axis. In this case, voltage (V, left y-axis) and current (uA, right y-axis).

(4.5.14) In the schematic editor window, replace the DC voltage source (**vdc**) with a time-varying Sinusoidal voltage source (**vsin**). You can use **Menu: Edit à Delete**, the '**Delete'** icon-button on the toolbar, or the '**delete'** key on the keyboard to delete the **vdc** part and add a new instance of the **vsin** part. Or you can change **vdc** to a **vsin** component directly by editing its properties (**Menu: Edit à Properties à Objects**, or **Edit Properties** icon-button, or press '**Q**') by changing the cell name from vdc to vsin. The 'Add Instance' form should then change from a vdc form to a vsin form.



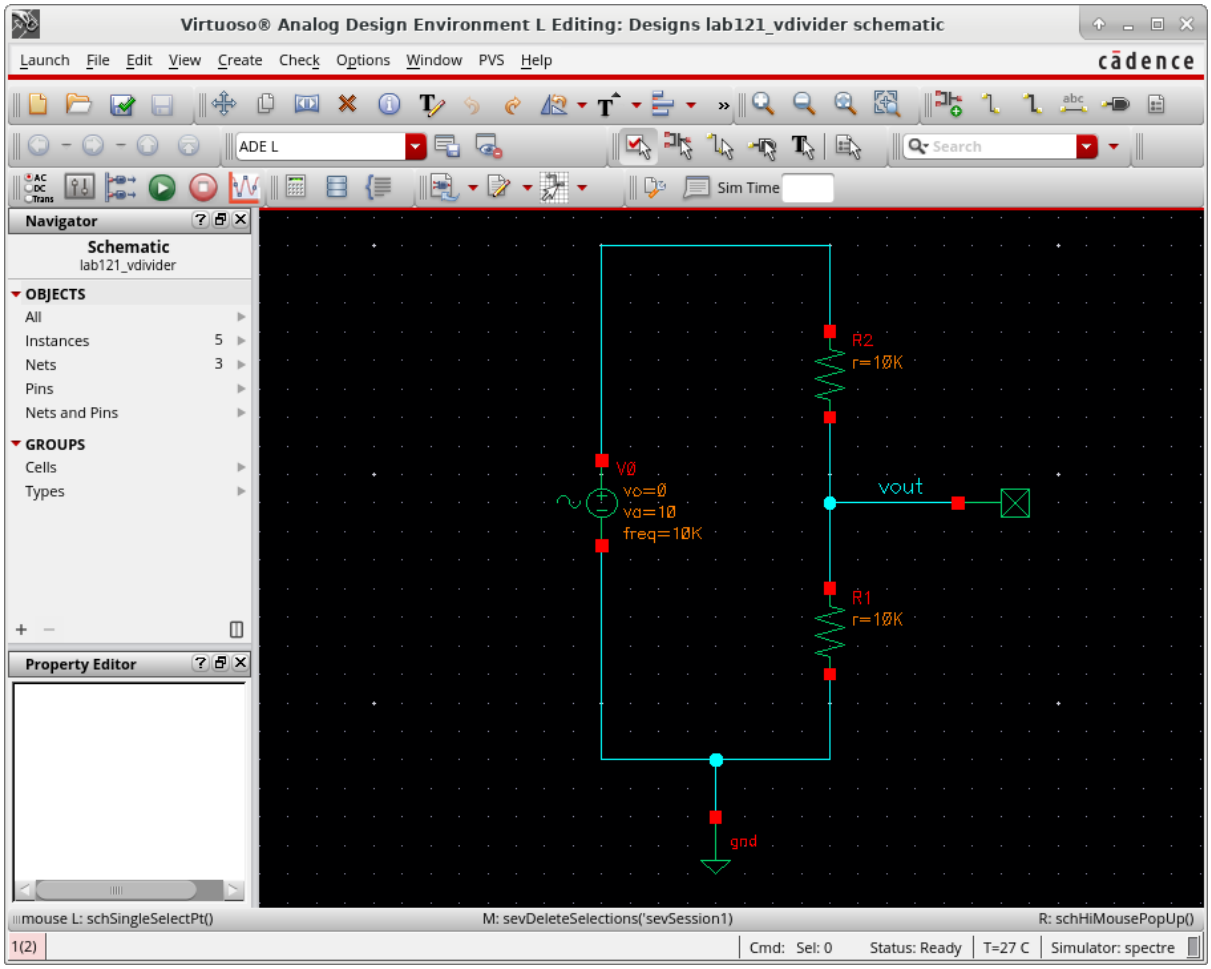
< Figure 23. The vsin Add Instance Form. >

(4.5.15) In the **vsin** instance form, set the attributes/parameters as follows: 'DC Voltage' = empty, 'Amplitude' = 10V, 'Offset' = 0V, and 'Frequency' 10K. See Figure 23. OK the form.

(4.5.16) Note: Most sources in a SPICE-based simulator, have several attributes for specifying a sinusoidal amplitude. Some specify the amplitude for use with a time-domain (transient) analysis, while others specify the amplitude for use with a small-signal (AC) analysis. In the case of the **vsin** component, make sure to set the 'Amplitude' attribute for the transient

analysis and not the 'AC Magnitude' attribute – these are not the same. The AC Magnitude is used for the various small-signal analyses of the ADE and is ignored by the transient analysis. (Similarly, the 'DC Voltage' attribute is used only by DC Analysis – it is ignored by the transient analysis. To give the source a DC bias use the 'Offset' attribute.)

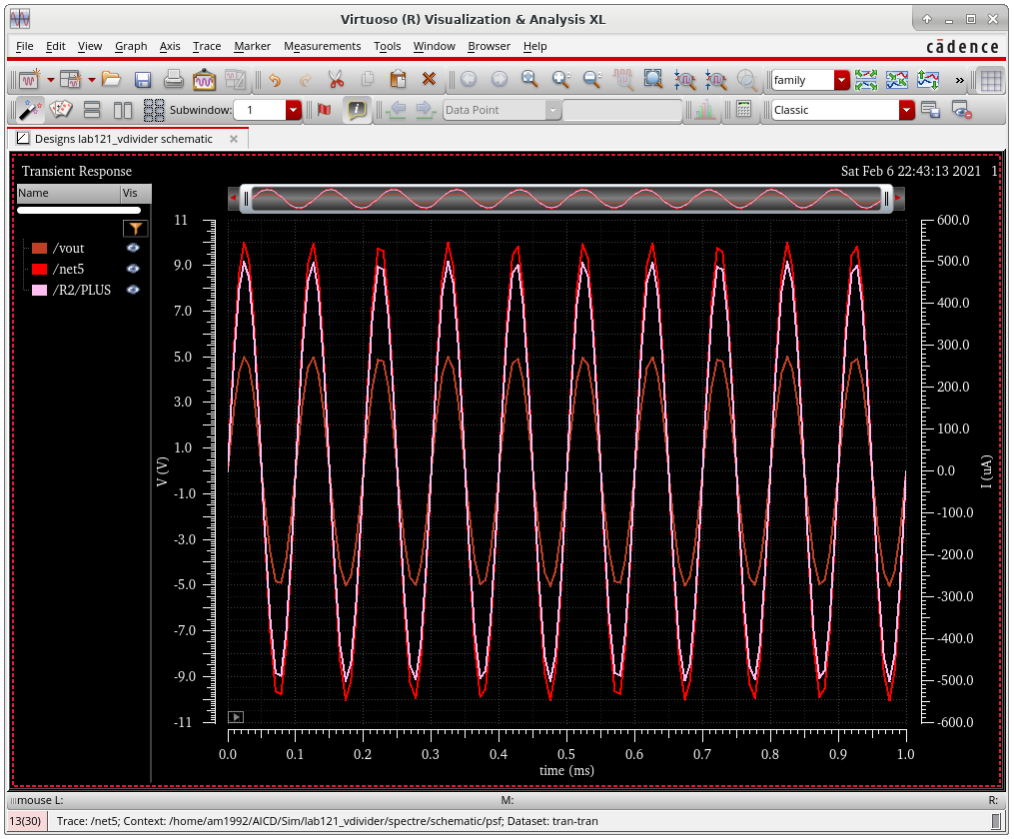
(4.5.17) **Check and Save** the schematic. There should be no errors or warnings. Use the CIW to diagnose any errors or warnings (if there are any).



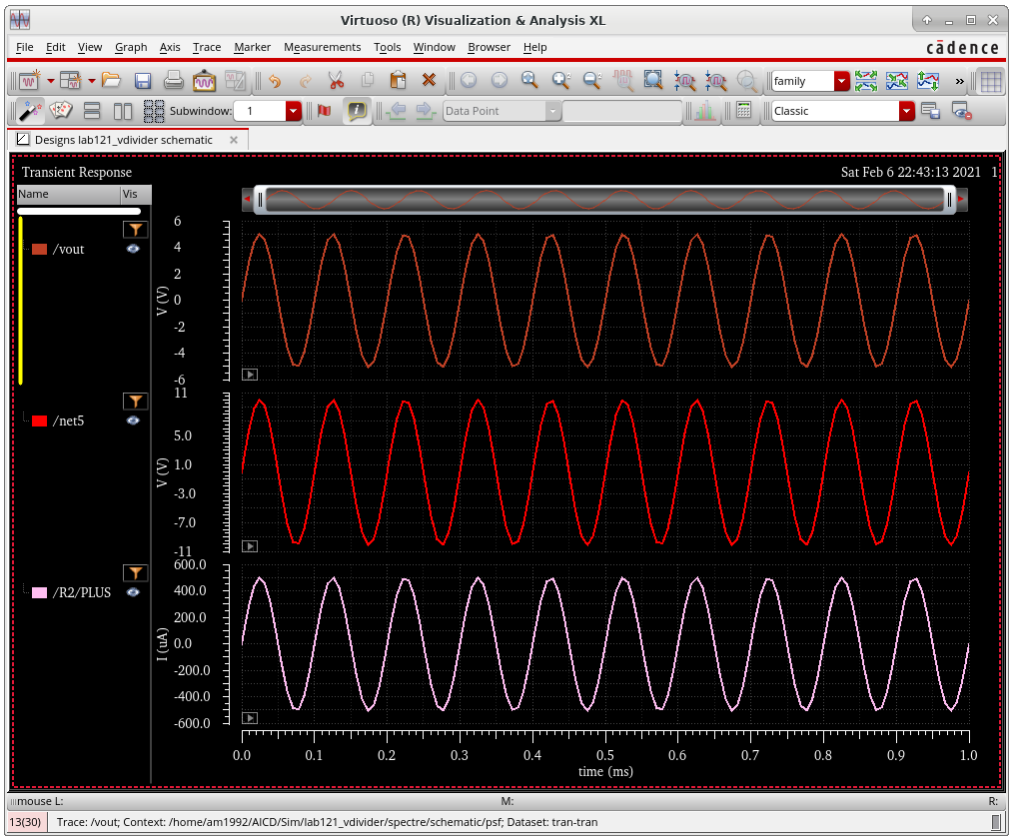
< Figure 24. The complete voltage divider circuit in the schematic editor with vsin source. Amplitude = 10V, Frequency = 10KHz, DC Offset = 0V. >

(4.5.18) We will now re-simulate the schematic. In the Analog Design Environment window, re-run the '**Netlist and Run**' command by clicking on the green play button. If the simulation runs properly, both the textual output window as well as the waveform output window will appear. See Figure 25. Note that this time the waveform varies with time over the duration of the analysis. The peak values of the voltage and current waveforms remain the same: supply net (=10V), output net (=5V), and node current (=500uA).

(4.5.19) You can separate the various plots in the waveform window to make it more readable. Right-click anywhere on the empty (black) space of the waveform window to invoke a context-menu, and select **Split Current Strip à trace**. Each trace will now be presented in its own space.



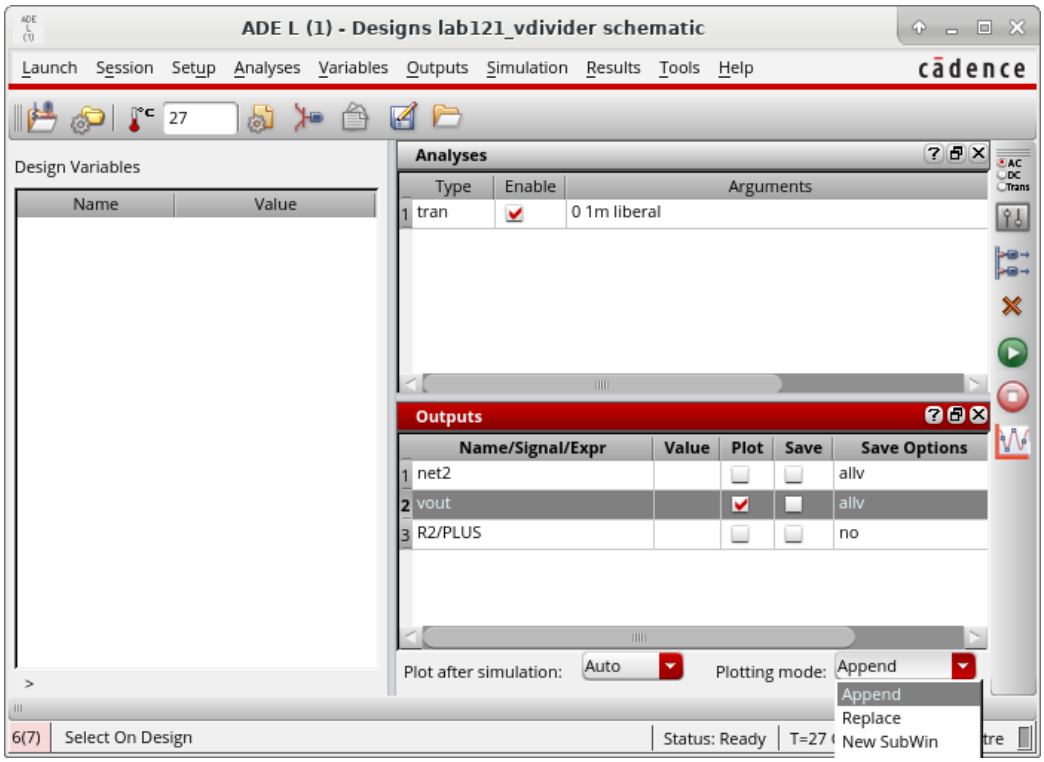
< Figure 25. The Waveform Output Window for voltage divider with vsin source. >



< Figure 26. The Waveform Output Window – with split traces. >

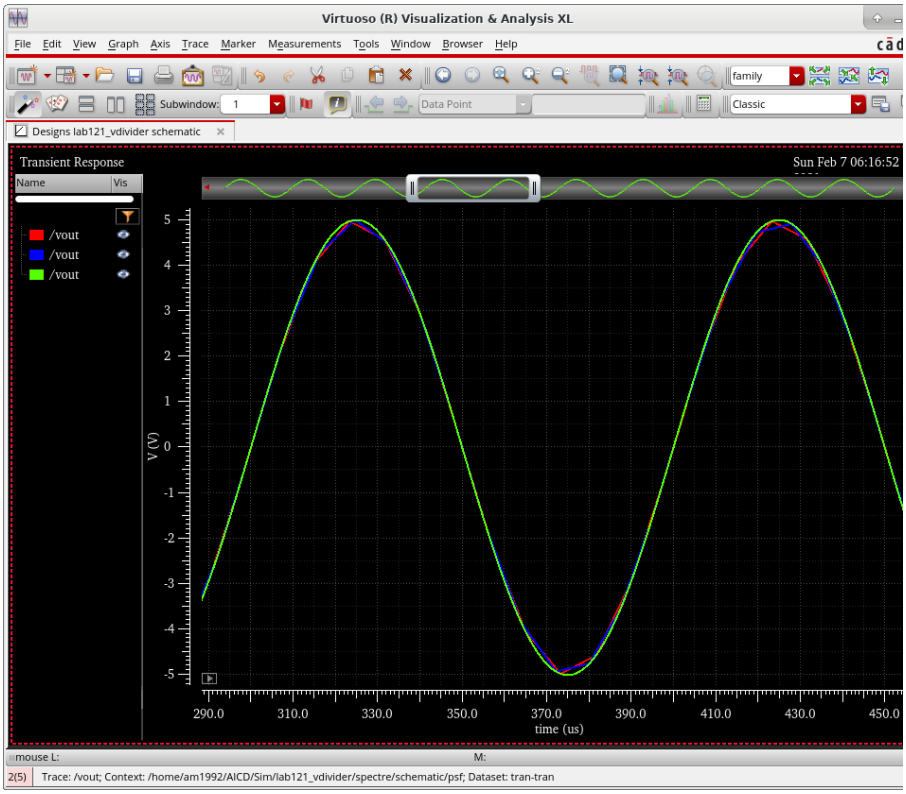
(4.5.20) Finally, re-run the transient analysis for the three different transient analysis accuracy options: liberal, moderate, and conservative. Observe, the output waveforms. Note that for the liberal option, the number of sample points that make-up the trace is fewer (i.e. larger time-step between solved points), leading to a piecewise linear (coarse, jagged) appearance. With the conservative option, the appearance of the trace is far smoother due to the high number of sample points (shorter time-step between solved points).

To plot all three waveforms in the same plot window, you will need to change the plotting mode of the ADE to 'Append'. On the bottom right corner of the ADE window there is a drop-down menu called '**Plotting mode**'. By default, this is set to replace; this means every time you run the simulation the contents of the plot will be overwritten with the new simulation results. By setting this to Append the previous simulation results are preserved in the plotting window and are displayed along the current simulation data. See Figure 27.

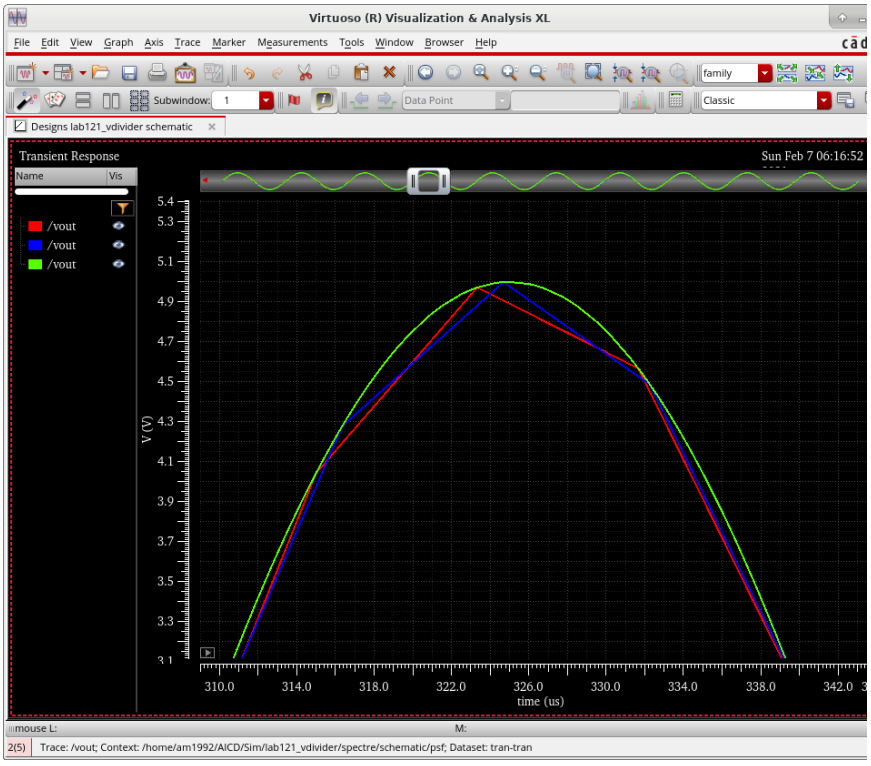


< Figure 277. ADE window showing plotting mode drop down menu (bottom-right) set to 'Append'. >

For a simple circuit like the voltage divider, the increase in simulation time is not noticeable so the trade-off between the liberal, moderate, and conservative accuracy setting is not immediately apparent. However, for larger and more complex circuits, a loss of accuracy for a faster simulation time may be worthwhile – at least during the early stages of the design flow where the focus may be on getting the design working rather than accuracy. In the latter stages of the design flow, high accuracy simulations may be preferred.



< Figure 288. The Waveform Output Window – traces with liberal (red), moderate (blue), and conservative (red) accuracy settings. >



< Figure 29. The Waveform Output Window – traces with liberal (red), moderate (blue), and conservative (red) accuracy settings. Close-up >

End of Lab!